Submitted to the Annals of Statistics

SUPPLEMENTARY MATERIAL FOR DISTRIBUTION AND CORRELATION FREE TWO-SAMPLE TEST OF HIGH DIMENSIONAL MEANS

BY KAIJIE XUE AND FANG YAO

Nankai University and Peking University

1. Notations. A brief review of some relevant notations is given in the following before the proofs. For two vectors $x = (x_1, \ldots, x_p) \in \mathbb{R}^p$ and $y = (y_1, \ldots, y_p)' \in \mathbb{R}^p$, write $x \leq y$ if $x_j \leq y_j$ for all $j = 1, \ldots, p$. For any $x = (x_1, \ldots, x_p)' \in \mathbb{R}^p$ and $a \in \mathbb{R}$, denote $x + a = (x_1 + a, \ldots, x_p + a)'$. For any $a, b \in \mathbb{R}$, use the notations $a \vee b = \max\{a, b\}$ and $a \wedge b = \min\{a, b\}$. For any two sequences of constants a_n and b_n, write $a_n \lesssim b_n$ if $a_n \leq C b_n$ up to a universal constant $C > 0$, and $a_n \sim b_n$ if $a_n \lesssim b_n$ and $b_n \lesssim a_n$. For any matrix $A = (a_{ij})$, define $\|A\|_\infty = \max_{i,j} |a_{ij}|$. For any function $f : \mathbb{R} \to \mathbb{R}$, write $\|f\|_\infty = \sup_{z \in \mathbb{R}} |f(z)|$. For a smooth function $g : \mathbb{R}^p \to \mathbb{R}$, we adopt indices to represent the partial derivatives for brevity, for example, $\partial_j \partial_k \partial_l g = g_{jkl}$. For any $\alpha > 0$, define the function $\psi_\alpha(x) = \exp(x^\alpha) - 1$ for $x \in [0, \infty)$, then for any random variable X, define

$$\|X\|_{\psi_\alpha} = \inf\{\lambda > 0 : E\{\psi_\alpha(|X|/\lambda)\} \leq 1\}, \tag{1}$$

which is an Orlicz norm for $\alpha \in [1, \infty)$ and a quasi-norm for $\alpha \in (0, 1)$. For any $\beta > 0$ and $y \in \mathbb{R}^p$, we define a function $F_\beta(w)$ as

$$F_\beta(w) = \beta^{-1} \log \big[\sum_{j=1}^{p} \exp\{\beta(w_j - y_j)\} \big], w \in \mathbb{R}^p, \tag{2}$$

which satisfies the property

$$0 \leq F_\beta(w) - \max_{1 \leq j \leq p} (w_j - y_j) \leq \beta^{-1} \log p, \tag{3}$$

for every $w \in \mathbb{R}^p$ by (1) in [2]. In addition, we let $\varphi_0 : \mathbb{R} \to [0, 1]$ be a real valued function such that φ_0 is thrice continuously differentiable and $\varphi_0(z) = 1$ for $z \leq 0$ and $\varphi_0(z) = 0$ for $z \geq 1$. For any $\phi \geq 1$, we define a function $\varphi(z) = \varphi_0(\phi z), z \in \mathbb{R}$. Then, for any $\phi \geq 1$ and $y \in \mathbb{R}^p$, we denote $\beta = \phi \log p$ and define a function $\kappa : \mathbb{R}^p \to [0, 1]$ as

$$\kappa(w) = \varphi_0(\phi F_{\phi \log p}(w)) = \varphi(F_\beta(w)), w \in \mathbb{R}^p. \tag{4}$$

2. Proofs of Theorems 1–2, and Corollary 1. *Proof of Theorem 1*: First, we set $\bar{L}_n^* = \bar{L}_m^{**} = B_{n,m}$. Together with condition (c), it is obvious that

$$\bar{L}_n^* = B_{n,m} \geq L_n^X, \quad \bar{L}_m^{**} = B_{n,m} \geq L_m^Y. \tag{5}$$

By combining the definition of the Orlicz norm in (1) with condition (d), we have

$$\max_{1\leq i\leq n}\max_{1\leq j\leq p}\|X_{ij}-\mu_j^X\|_{\psi_1} \leq B_{n,m}, \max_{1\leq i\leq m}\max_{1\leq j\leq p}\|Y_{ij}-\mu_j^Y\|_{\psi_1} \leq B_{n,m}. \tag{6}$$

Moreover, since $F_{ij} \sim N(\mu_j^X, E(X_{ij}-\mu_j^X)^2)$ for all $i = 1,\ldots,n$ and $j = 1,\ldots,p$, it can be deduced that for any $t \geq 0$,

$$P(|F_{ij}-\mu_j^X| > t) \leq 2\exp\big[-t^2/\{2E(X_{ij}-\mu_j^X)^2\}\big], \tag{7}$$

for all $i = 1,\ldots,n$ and $j = 1,\ldots,p$. Therefore, by combining (7) with Lemma 2.2.1 in [4], we have

$$\|F_{ij}-\mu_j^X\|_{\psi_2} \leq \big[(1+2)\{2E(X_{ij}-\mu_j^X)^2\}\big]^{1/2}, \tag{8}$$

for all $i = 1,\ldots,n$ and $j = 1,\ldots,p$. Furthermore, by property of the Orlicz norm, it can be shown that for all $i = 1,\ldots,n$ and $j = 1,\ldots,p$,

$$\|F_{ij}-\mu_j^X\|_{\psi_1} \leq (\log 2)^{2^{-1}-1}\|F_{ij}-\mu_j^X\|_{\psi_2} = (\log 2)^{-2^{-1}}\|F_{ij}-\mu_j^X\|_{\psi_2}. \tag{9}$$

In addition, for all $i = 1,\ldots,n$ and $j = 1,\ldots,p$, it can be justified that

$$E(X_{ij}-\mu_j^X)^2 \leq 2\|X_{ij}-\mu_j^X\|_{\psi_1}^2. \tag{10}$$

Henceforth, by combining (6), (8), (9) with (10), we have

$$\max_{1\leq i\leq n}\max_{1\leq j\leq p}\|F_{ij}-\mu_j^X\|_{\psi_1} \leq c_2 B_{n,m}, \tag{11}$$

for a universal constant $c_2 > 0$. Similar reasoning leads to

$$\max_{1\leq i\leq m}\max_{1\leq j\leq p}\|G_{ij}-\mu_j^Y\|_{\psi_1} \leq c_3 B_{n,m}, \tag{12}$$

for a universal constant $c_3 > 0$. As a result, by combining (6), (11), (12) with Lemma 2.2.2 in [4], we conclude that

$$\begin{aligned}
&\|\max_{1\leq j\leq p}|X_{ij}-\mu_j^X|\|_{\psi_1} \leq c_4 B_{n,m}\log p, \quad i = 1,\ldots,n,\\
&\|\max_{1\leq j\leq p}|F_{ij}-\mu_j^X|\|_{\psi_1} \leq c_5 B_{n,m}\log p, \quad i = 1,\ldots,n,\\
&\|\max_{1\leq j\leq p}|Y_{ij}-\mu_j^Y|\|_{\psi_1} \leq c_6 B_{n,m}\log p, \quad i = 1,\ldots,m,\\
&\|\max_{1\leq j\leq p}|G_{ij}-\mu_j^Y|\|_{\psi_1} \leq c_7 B_{n,m}\log p, \quad i = 1,\ldots,m,
\end{aligned} \tag{13}$$

where c_4, c_5, c_6 and c_7 are some positive universal constants. It then follows from (13) and Markov's inequality that for any $t \geq 0$,

$$
\begin{aligned}
&P(\max_{1\leq j\leq p} |X_{ij} - \mu_j^X| > t) \leq 2\exp\{-t/(c_4 B_{n,m}\log p)\}, \quad i = 1,\ldots,n,\\
&P(\max_{1\leq j\leq p} |F_{ij} - \mu_j^X| > t) \leq 2\exp\{-t/(c_5 B_{n,m}\log p)\}, \quad i = 1,\ldots,n,\\
&P(\max_{1\leq j\leq p} |Y_{ij} - \mu_j^Y| > t) \leq 2\exp\{-t/(c_6 B_{n,m}\log p)\}, \quad i = 1,\ldots,m,\\
&P(\max_{1\leq j\leq p} |G_{ij} - \mu_j^Y| > t) \leq 2\exp\{-t/(c_7 B_{n,m}\log p)\}, \quad i = 1,\ldots,m. \qquad (14)
\end{aligned}
$$

Putting condition (b) and Lemma 3 together yields that for any $\phi_1, \phi_2 \geq 1$,

$$
\begin{aligned}
\rho_{n,m}^* \leq & K^*\Big[n^{-1/2}\phi_1^2(\log p)^2\big\{\phi_1 L_n^X \rho_{n,m}^* + L_n^X(\log p)^{1/2} + \phi_1 M_n(\phi_1)\big\} + \\
& m^{-1/2}\phi_2^2(\log p)^2|\delta_{n,m}|^3\big\{\phi_2 L_m^Y \rho_{n,m}^* + L_m^Y(\log p)^{1/2} + \phi_2 M_m^*(\phi_2)\big\} + \\
& (\min\{\phi_1,\phi_2\})^{-1}(\log p)^{1/2}\Big],
\end{aligned}
$$

for a universal constant $K^* > 0$ that depends only on b, where $\rho_{n,m}^*$ is defined in (11) from the main article. Accordingly, we set a universal constant

$$
K_1 = \min\{8^{-1}c_4^{-1}c_1^{-1/3}, 8^{-1}c_5^{-1}c_1^{-1/3}, 8^{-1}c_6^{-1}c_1^{-1/3}, 8^{-1}c_7^{-1}c_1^{-1/3}, (K^* \vee 2)^{-1}\}, \qquad (15)
$$

and it is trivial to check that

$$
K_1 \in (0, (K^* \vee 2)^{-1}]. \qquad (16)
$$

Then, we let

$$
\begin{aligned}
\phi_n^* =& K_1\{(B_{n,m})^2(\log p)^4/n\}^{-1/6}, \qquad (17)\\
\phi_m^{**} =& K_1\{(B_{n,m})^2(\log p)^4|\delta_{n,m}|^6/m\}^{-1/6}.
\end{aligned}
$$

Together with conditions (a) and (e), it can be verified that

$$
\phi_n^* \geq 2, \quad \phi_m^{**} \geq 2. \qquad (18)
$$

To this end, by combining (14), (17) with Lemma C.1 in [3], we reach a conclusion that

$$
\begin{aligned}
M_n^X(\phi_n^*) \lesssim & \big\{n^{1/2}/(\phi_n^*\log p) + B_{n,m}\log p\big\}^3 \exp\big[-n^{1/2}/\{4c_4\phi_n^* B_{n,m}(\log p)^2\}\big],\\
M_n^F(\phi_n^*) \lesssim & \big\{n^{1/2}/(\phi_n^*\log p) + B_{n,m}\log p\big\}^3 \exp\big[-n^{1/2}/\{4c_5\phi_n^* B_{n,m}(\log p)^2\}\big],\\
M_m^Y(\phi_m^{**}) \lesssim & \big\{m^{1/2}/(|\delta_{n,m}|\phi_m^{**}\log p) + B_{n,m}\log p\big\}^3\\
& \exp\big[-m^{1/2}/\{4c_6|\delta_{n,m}|\phi_m^{**} B_{n,m}(\log p)^2\}\big],\\
M_m^G(\phi_m^{**}) \lesssim & \big\{m^{1/2}/(|\delta_{n,m}|\phi_m^{**}\log p) + B_{n,m}\log p\big\}^3\\
& \exp\big[-m^{1/2}/\{4c_7|\delta_{n,m}|\phi_m^{**} B_{n,m}(\log p)^2\}\big]. \qquad (19)
\end{aligned}
$$

To bound the term $M_n^X(\phi_n^*)$, it can be noted that

$$\begin{aligned} & n^{1/2}/\{4c_4\phi_n^* B_{n,m}(\log p)^2\} \\ \geq & (4c_4K_1)^{-1}\left[(B_{n,m})^2\{\log(pn)\}^7/n\right]^{-1/3}\log(pn) \\ \geq & (4c_4K_1)^{-1}c_1^{-1/3}\log(pn) = K_1^{-1}4^{-1}c_4^{-1}c_1^{-1/3}\log(pn) \\ \geq & (8^{-1}c_4^{-1}c_1^{-1/3})^{-1}4^{-1}c_4^{-1}c_1^{-1/3}\log(pn) = 2\log(pn), \end{aligned} \tag{20}$$

where the second inequality follows from condition (e), and the last inequality is by (15). Moreover, we have that

$$n^{1/2}/(\phi_n^*\log p) \leq n^{1/2}/(2\log p) \lesssim n^{1/2}, \tag{21}$$

where the first inequality holds true due to (18). It can also be checked that

$$B_{n,m}\log p = \{(B_{n,m})^2(\log p)^2/n\}^{1/2}n^{1/2} \lesssim n^{1/2}, \tag{22}$$

where the last inequality is valid according to condition (e). At this point, by combining (19), (20), (21) with (22), it can be obtained that

$$M_n^X(\phi_n^*) \lesssim n^{3/2}(pn)^{-2} \lesssim n^{-1/2}. \tag{23}$$

Similar argument also gives

$$M_n^F(\phi_n^*) \lesssim n^{-1/2}. \tag{24}$$

Piecing (23) and (24) together leads to

$$M_n(\phi_n^*) = M_n^X(\phi_n^*) + M_n^F(\phi_n^*) \lesssim n^{-1/2}. \tag{25}$$

To bound the term $M_m^Y(\phi_m^{**})$, notice that

$$\begin{aligned} & m^{1/2}/\{4c_6|\delta_{n,m}|\phi_m^{**}B_{n,m}(\log p)^2\} \\ \geq & (4c_6K_1)^{-1}\left[(B_{n,m})^2\{\log(pm)\}^7/m\right]^{-1/3}\log(pm) \\ \geq & (4c_6K_1)^{-1}c_1^{-1/3}\log(pm) = K_1^{-1}4^{-1}c_6^{-1}c_1^{-1/3}\log(pm) \\ \geq & (8^{-1}c_6^{-1}c_1^{-1/3})^{-1}4^{-1}c_6^{-1}c_1^{-1/3}\log(pm) = 2\log(pm), \end{aligned} \tag{26}$$

where the second inequality is based on condition (e), and the last inequality is owing to (15). Also notice that

$$m^{1/2}/(|\delta_{n,m}|\phi_m^{**}\log p) \lesssim m^{1/2}/(\phi_m^{**}\log p) \lesssim m^{1/2}/(2\log p) \lesssim m^{1/2}, \tag{27}$$

where the first inequality follows from condition (a), and the second inequality is based on (18). In addition, one has

$$B_{n,m}\log p = \{(B_{n,m})^2(\log p)^2/m\}^{1/2}m^{1/2} \lesssim m^{1/2}, \tag{28}$$

where the last inequality follows from condition (e). Therefore, by combining (19), (26), (27) with (28), we have

$$M_m^Y(\phi_m^{**}) \lesssim m^{3/2}(pm)^{-2} \lesssim m^{-1/2}. \tag{29}$$

Similar argument yields that

$$M_m^G(\phi_m^{**}) \lesssim m^{-1/2}. \tag{30}$$

Under (29) and (30), we have

$$M_m^*(\phi_m^{**}) = M_m^Y(\phi_m^{**}) + M_m^G(\phi_m^{**}) \lesssim m^{-1/2}. \tag{31}$$

To this end, by combining condition (a), condition (b), (5), (16), (17), (18), (25), (31) with Lemma 4, we conclude that

$$\begin{aligned}\rho_{n,m}^{**} \lesssim & \{(B_{n,m})^2(\log p)^7/n\}^{1/6} + n^{-1/2}(B_{n,m})^{-1} + \\ & \{(B_{n,m})^2(\log p)^7|\delta_{n,m}|^6/m\}^{1/6} + m^{-1/2}(B_{n,m})^{-1} \\ \lesssim & \{(B_{n,m})^2(\log p)^7/n\}^{1/6} + n^{-1/2}(B_{n,m})^{-1} + \{(B_{n,m})^2(\log p)^7/m\}^{1/6} \\ & + m^{-1/2}(B_{n,m})^{-1}.\end{aligned} \tag{32}$$

For the term $n^{-1/2}(B_{n,m})^{-1}$ in (32), it can be shown that

$$\begin{aligned}n^{-1/2}(B_{n,m})^{-1} &= \left[(B_{n,m})^2\{\log(pn)\}^7/n\right]^{1/6}(B_{n,m})^{-4/3}\{\log(pn)\}^{-7/6}n^{-1/3} \\ &\lesssim \left[(B_{n,m})^2\{\log(pn)\}^7/n\right]^{1/6}.\end{aligned} \tag{33}$$

Likewise, we have

$$m^{-1/2}(B_{n,m})^{-1} \lesssim \left[(B_{n,m})^2\{\log(pm)\}^7/m\right]^{1/6}. \tag{34}$$

Finally, based on (32), (33) and (34), one has

$$\rho_{n,m}^{**} \lesssim \left[(B_{n,m})^2\{\log(pn)\}^7/n\right]^{1/6} + \left[(B_{n,m})^2\{\log(pm)\}^7/m\right]^{1/6},$$

which completes the proof. □

Proof of Theorem 2: First, by combining (b) with Lemma 5, it is clear that for any sequence of constants $\bar{\Delta}_{n,m} > 0$, on the event $\{\hat{\Delta}_{n,m} \le \bar{\Delta}_{n,m}\}$, we have

$$\rho_{n,m}^{MB} \lesssim (\bar{\Delta}_{n,m})^{1/3}(\log p)^{2/3}. \tag{35}$$

Then, our goal is to find a proper bound for the quantity $\hat{\Delta}_{n,m} = \|\hat{\Sigma}^X - \Sigma^X + \delta_{n,m}^2(\hat{\Sigma}^Y - \Sigma^Y)\|_\infty$. To achieve this, we first note that

$$\hat{\Delta}_{n,m} \le \hat{\Delta}_{n,m}^{(1)} + (\hat{\Delta}_{n,m}^{(2)})^2 + \delta_{n,m}^2\hat{\Delta}_{n,m}^{(3)} + \delta_{n,m}^2(\hat{\Delta}_{n,m}^{(4)})^2, \tag{36}$$

where

$$\hat{\Delta}_{n,m}^{(1)} = \max_{1\le j,k\le p} |n^{-1}\sum_{i=1}^{n}\{(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X) - E(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)\}|,$$

$$\hat{\Delta}_{n,m}^{(2)} = \|(\bar{X}-\mu^X)(\bar{X}-\mu^X)'\|_\infty^{1/2} = \|\bar{X}-\mu^X\|_\infty = \max_{1\le j\le p}|n^{-1}\sum_{i=1}^{n}(X_{ij}-\mu_j^X)|,$$

$$\hat{\Delta}_{n,m}^{(3)} = \max_{1\le j,k\le p} |m^{-1}\sum_{i=1}^{m}\{(Y_{ij}-\mu_j^Y)(Y_{ik}-\mu_k^Y) - E(Y_{ij}-\mu_j^Y)(Y_{ik}-\mu_k^Y)\}|,$$

$$\hat{\Delta}_{n,m}^{(4)} = \|(\bar{Y}-\mu^Y)(\bar{Y}-\mu^Y)'\|_\infty^{1/2} = \|\bar{Y}-\mu^Y\|_\infty = \max_{1\le j\le p}|m^{-1}\sum_{i=1}^{m}(Y_{ij}-\mu_j^Y)|.$$

To bound the term $\hat{\Delta}_{n,m}^{(1)}$, we first denote

$$\sigma_{n,m,1}^2 = \max_{1\le j,k\le p}\sum_{i=1}^{n} E\{(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X) - E(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)\}^2,$$

and it can be shown that

$$\begin{aligned}
\sigma_{n,m,1}^2 &\le \max_{1\le j,k\le p}\sum_{i=1}^{n} E\{(X_{ij}-\mu_j^X)^2(X_{ik}-\mu_k^X)^2\} \\
&\le \max_{1\le j,k\le p}\Big\{\sum_{i=1}^{n} E(X_{ij}-\mu_j^X)^4\Big\}^{1/2}\Big\{\sum_{i=1}^{n} E(X_{ik}-\mu_k^X)^4\Big\}^{1/2} \\
&= \max_{1\le j\le p}\sum_{i=1}^{n} E(X_{ij}-\mu_j^X)^4 \le n(B_{n,m})^2, \qquad (37)
\end{aligned}$$

where the last inequality follows from condition (c). Moreover, we also denote

$$M_{n,m,1} = \max_{1\le i\le n}\max_{1\le j\le p}\max_{1\le k\le p}|(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X) - E(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)|,$$

and one can show that

$$
\begin{aligned}
\|M_{n,m,1}\|_{\psi_{1/2}} \leq & \| \max_{1\leq i\leq n} \max_{1\leq j,k\leq p} |(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)| + \\
& \max_{1\leq i\leq n} \max_{1\leq j,k\leq p} E\{|(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)|\}\|_{\psi_{1/2}} \\
\leq & c_1 \| \max_{1\leq i\leq n} \max_{1\leq j,k\leq p} |(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)|\|_{\psi_{1/2}} + \\
& c_1 \| \max_{1\leq i\leq n} \max_{1\leq j,k\leq p} E\{|(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)|\}\|_{\psi_{1/2}} \\
\leq & c_1 \| \max_{1\leq i\leq n} \max_{1\leq j,k\leq p} |(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)|\|_{\psi_{1/2}} + \\
& c_2 \max_{1\leq i\leq n} \max_{1\leq j,k\leq p} E\{|(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)|\} \\
= & c_1 \| \max_{1\leq i\leq n} \max_{1\leq j\leq p} |X_{ij}-\mu_j^X|\|_{\psi_1}^2 + \\
& c_2 \max_{1\leq i\leq n} \max_{1\leq j,k\leq p} E\{|(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)|\} \\
\leq & c_3 \big\{ \log(pn) \max_{1\leq i\leq n} \max_{1\leq j\leq p} \|X_{ij}-\mu_j^X\|_{\psi_1} \big\}^2 + \\
& c_2 \max_{1\leq i\leq n} \max_{1\leq j,k\leq p} E\{|(X_{ij}-\mu_j^X)(X_{ik}-\mu_k^X)|\} \\
\leq & c_3\{\log(pn)\}^2 (B_{n,m})^2 + c_2 \max_{1\leq i\leq n} \max_{1\leq j\leq p} E\{(X_{ij}-\mu_j^X)^2\} \\
\leq & c_3\{\log(pn)\}^2 (B_{n,m})^2 + 2c_2 \max_{1\leq i\leq n} \max_{1\leq j\leq p} \|X_{ij}-\mu_j^X\|_{\psi_1}^2 \\
\leq & c_3\{\log(pn)\}^2 (B_{n,m})^2 + 2c_2 (B_{n,m})^2 \leq c_4\{\log(pn)\}^2 (B_{n,m})^2, \qquad (38)
\end{aligned}
$$

where c_1, c_2, c_3 and c_4 are some positive universal constants, the second inequality is based on the fact that $\|\cdot\|_{\psi_{1/2}}$ is a quasi-norm, the fourth inequality follows from Lemma 2.2.2 in [4], and the fifth inequality is based on condition (d). Furthermore, we have

$$
\{E(M_{n,m,1}^2)\}^{1/2} \leq (4!\|M_{n,m,1}\|_{\psi_{1/2}}^2)^{1/2} \leq c_5\{\log(pn)\}^2 (B_{n,m})^2, \qquad (39)
$$

where $c_5 > 0$ is a universal constant, and the last inequality is based on (38). To this end, by combining (37), (39) with Lemma E.1 in [3], we have

$$
\begin{aligned}
E(\hat{\Delta}_{n,m}^{(1)}) \leq & c_6 n^{-1} \big[(\sigma_{n,m,1}^2 \log p)^{1/2} + \{E(M_{n,m,1}^2)\}^{1/2} \log p \big] \\
\leq & c_6 n^{-1} \big[(nB_{n,m}^2 \log p)^{1/2} + c_5\{\log(pn)\}^2 (B_{n,m})^2 \log p \big] \\
\leq & c_6 n^{-1} \big[\{nB_{n,m}^2 \log(pn)\}^{1/2} + c_5\{\log(pn)\}^3 (B_{n,m})^2 \big] \\
\leq & c_6(1+c_5)\{n^{-1}B_{n,m}^2 \log(pn)\}^{1/2} \leq c_7\{n^{-1}B_{n,m}^2 \log(pn)\}^{1/2}, \qquad (40)
\end{aligned}
$$

where c_6 and c_7 are some positive universal constants, and the fourth inequality is based on condition (e). Therefore, by combining (37), (38), (40) with Lemma E.2 in [3], it can be deduced that for any $t > 0$,

$$\begin{aligned}
&P\big[\hat{\Delta}^{(1)}_{n,m} > 2c_7\{n^{-1}B^2_{n,m}\log(pn)\}^{1/2} + t\big] \\
\le& P\big\{\hat{\Delta}^{(1)}_{n,m} \ge 2E(\hat{\Delta}^{(1)}_{n,m}) + t\big\} \\
\le& \exp\big\{-n^2t^2/(3\sigma^2_{n,m,1})\big\} + 3\exp\big[-\big\{nt/(c_8\|M_{n,m,1}\|_{\psi_{1/2}})\big\}^{1/2}\big] \\
\le& \exp\big\{-nt^2/(3B^2_{n,m})\big\} + 3\exp\big[-\big\{nt/(c_8\|M_{n,m,1}\|_{\psi_{1/2}})\big\}^{1/2}\big] \\
\le& \exp\big\{-nt^2/(3B^2_{n,m})\big\} + 3\exp\big[-(nt)^{1/2}/\{c_9B_{n,m}\log(pn)\}\big], \qquad (41)
\end{aligned}$$

where c_8 and c_9 are some positive universal constants, the first inequality is by (40), the second inequality is based on Lemma E.2 in [3], the third inequality follows from (37), and the fourth inequality is due to (38). By substituting $t = \{n^{-1}B^2_{n,m}\log(pn)\}^{1/2}\log(1/\alpha_{n,m})$ into (41), it can be obtained that

$$\begin{aligned}
&P\big[\hat{\Delta}^{(1)}_{n,m} > 2c_7\{n^{-1}B^2_{n,m}\log(pn)\}^{1/2} + \{n^{-1}B^2_{n,m}\log(pn)\}^{1/2}\log(1/\alpha_{n,m})\big] \\
\le& \exp\big\{-\log(pn)\log^2(1/\alpha_{n,m})/3\big\} + \\
&3\exp\big[-\{B^2_{n,m}\log^3(pn)\log^{-2}(1/\alpha_{n,m})/n\}^{-1/4}/c_9\big] \\
\le& (\alpha_{n,m})^{\log(pn)/3} + 3(\alpha_{n,m})^{\log^{1/2}(pn)/c_9}.
\end{aligned}$$

Together with the fact that

$$\begin{aligned}
&2c_7\{n^{-1}B^2_{n,m}\log(pn)\}^{1/2} + \{n^{-1}B^2_{n,m}\log(pn)\}^{1/2}\log(1/\alpha_{n,m}) \\
\le& c_{10}\{n^{-1}B^2_{n,m}\log(pn)\}^{1/2}\log(1/\alpha_{n,m})
\end{aligned}$$

for a universal constant $c_{10} > 0$, we conclude that

$$\begin{aligned}
&P\big[\hat{\Delta}^{(1)}_{n,m} > c_{10}\{B^2_{n,m}\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/2}\big] \\
\le& (\alpha_{n,m})^{\log(pn)/3} + 3(\alpha_{n,m})^{\log^{1/2}(pn)/c_9}. \qquad (42)
\end{aligned}$$

Similar reasoning leads to

$$\begin{aligned}
&P\big[\hat{\Delta}^{(3)}_{n,m} > c_{11}\{B^2_{n,m}\log(pm)\log^2(1/\alpha_{n,m})/m\}^{1/2}\big] \\
\le& (\alpha_{n,m})^{\log(pm)/3} + 3(\alpha_{n,m})^{\log^{1/2}(pm)/c_{12}}, \qquad (43)
\end{aligned}$$

for some universal constants $c_{11}, c_{12} > 0$. Next, to bound the term $\hat{\Delta}^{(2)}_{n,m}$, we first denote $\sigma^2_{n,m,2} = \max_{1\le j\le p}\sum_{i=1}^n E\{(X_{ij} - \mu^X_j)^2\}$, and one can deduce that

$$\sigma^2_{n,m,2} \le 2\max_{1\le j\le p}\sum_{i=1}^n \|X_{ij} - \mu^X_j\|^2_{\psi_1} \le 2nB^2_{n,m}, \qquad (44)$$

where the last inequality is by condition (d). In addition, we also denote $M_{n,m,2} = \max_{1\le i\le n}\max_{1\le j\le p}|X_{ij}-\mu_j^X|$, and one can show that

$$\|M_{n,m,2}\|_{\psi_1} \le c_{13}\log(pn)\max_{1\le i\le n}\max_{1\le j\le p}\|X_{ij}-\mu_j^X\|_{\psi_1} \le c_{13}\log(pn)B_{n,m}, \tag{45}$$

for a universal constant $c_{13}>0$, which further implies that

$$\{E(M_{n,m,2}^2)\}^{1/2} \le (2!\|M_{n,m,2}\|_{\psi_1}^2)^{1/2} \le c_{14}\log(pn)B_{n,m}, \tag{46}$$

where $c_{14}>0$ is a universal constant. Hence, by combining (44), (46) with Lemma E.1 in [3], we have

$$\begin{aligned} E(\hat\Delta_{n,m}^{(2)}) \le& c_{15}n^{-1}\big[(\sigma_{n,m,2}^2\log p)^{1/2} + \{E(M_{n,m,2}^2)\}^{1/2}\log p\big] \\ \le& c_{15}n^{-1}\big[\{2nB_{n,m}^2\log(pn)\}^{1/2} + c_{14}B_{n,m}\log^2(pn)\big] \\ \le& c_{15}\{B_{n,m}^2\log(pn)/n\}^{1/2}\big[2^{1/2} + c_{14}\{\log^3(pn)/n\}^{1/2}\big] \\ \le& c_{16}\{B_{n,m}^2\log(pn)/n\}^{1/2}, \end{aligned} \tag{47}$$

where c_{15} and c_{16} are positive universal constants. Then, by combining (44), (45), (47) with Lemma E.2 in [3], we have that for any $t>0$,

$$\begin{aligned} &P\big[\hat\Delta_{n,m}^{(2)} > 2c_{16}\{B_{n,m}^2\log(pn)/n\}^{1/2} + t\big] \le P\big\{\hat\Delta_{n,m}^{(2)} \ge 2E(\hat\Delta_{n,m}^{(2)}) + t\big\} \\ &\le \exp\big\{-n^2t^2/(3\sigma_{n,m,2}^2)\big\} + 3\exp\big\{-nt/(c_{17}\|M_{n,m,2}\|_{\psi_1})\big\} \\ &\le \exp\big\{-nt^2/(6B_{n,m}^2)\big\} + 3\exp\big[-nt/\{c_{18}B_{n,m}\log(pn)\}\big], \end{aligned} \tag{48}$$

where c_{17} and c_{18} are some positive universal constants. Likewise, we substitute $t = \{B_{n,m}^2\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/4}$ into (48) to obtain

$$\begin{aligned} &P\big[\hat\Delta_{n,m}^{(2)} > 2c_{16}\{B_{n,m}^2\log(pn)/n\}^{1/2} + \{B_{n,m}^2\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/4}\big] \\ &\le \exp\big\{-n^{1/2}\log^{1/2}(pn)\log(1/\alpha_{n,m})/(6B_{n,m})\big\} + \\ &\quad 3\exp\big[-n^{3/4}\log^{1/2}(1/\alpha_{n,m})/\{c_{18}B_{n,m}^{1/2}\log^{3/4}(pn)\}\big] \\ &\le \exp\big\{-\log^3(pn)\log(1/\alpha_{n,m})/6\big\} + 3\exp\big\{-\log^3(pn)\log(1/\alpha_{n,m})/c_{18}\big\} \\ &= (\alpha_{n,m})^{\log^3(pn)/6} + 3(\alpha_{n,m})^{\log^3(pn)/c_{18}}. \end{aligned} \tag{49}$$

Moreover, we have

$$\begin{aligned} &2c_{16}\{B_{n,m}^2\log(pn)/n\}^{1/2} + \{B_{n,m}^2\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/4} \\ \le& \{B_{n,m}^2\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/4}\{1 + 2c_{16}\log^{-1}(pn)\} \\ \le& c_{19}\{B_{n,m}^2\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/4}, \end{aligned} \tag{50}$$

for a universal constant $c_{19} > 0$. Therefore, by combining (49) with (50), we conclude that

$$\begin{aligned} &P\big[\hat{\Delta}^{(2)}_{n,m} > c_{19}\{B^2_{n,m}\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/4}\big] \\ \leq&(\alpha_{n,m})^{\log^3(pn)/6} + 3(\alpha_{n,m})^{\log^3(pn)/c_{18}}. \end{aligned} \tag{51}$$

Similar reasoning leads to

$$\begin{aligned} &P\big[\hat{\Delta}^{(4)}_{n,m} > c_{20}\{B^2_{n,m}\log(pm)\log^2(1/\alpha_{n,m})/m\}^{1/4}\big] \\ \leq&(\alpha_{n,m})^{\log^3(pm)/6} + 3(\alpha_{n,m})^{\log^3(pm)/c_{21}}, \end{aligned} \tag{52}$$

for some universal constants $c_{20}, c_{21} > 0$. To this end, we set $\bar{\Delta}_{n,m}$ to be

$$\begin{aligned} \bar{\Delta}_{n,m} =&(c_{10} + c^2_{19})\{B^2_{n,m}\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/2}+ \\ &(c_{11}\delta^2_1 + c^2_{20}\delta^2_1)\{B^2_{n,m}\log(pm)\log^2(1/\alpha_{n,m})/m\}^{1/2}, \end{aligned} \tag{53}$$

where δ_1 is defined in condition (a). Henceforth, by combining (36), (42), (43), (51), (52) with (53), we reach a conclusion that

$$\begin{aligned} &P(\hat{\Delta}_{n,m} \leq \bar{\Delta}_{n,m}) = 1 - P(\hat{\Delta}_{n,m} > \bar{\Delta}_{n,m}) \\ =&1 - P\{\hat{\Delta}^{(1)}_{n,m} + (\hat{\Delta}^{(2)}_{n,m})^2 + \delta^2_{n,m}\hat{\Delta}^{(3)}_{n,m} + \delta^2_{n,m}(\hat{\Delta}^{(4)}_{n,m})^2 > \bar{\Delta}_{n,m}\} \\ \geq&1 - P\big[\hat{\Delta}^{(1)}_{n,m} > c_{10}\{B^2_{n,m}\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/2}\big] \\ &- P\big[\hat{\Delta}^{(2)}_{n,m} > c_{19}\{B^2_{n,m}\log(pn)\log^2(1/\alpha_{n,m})/n\}^{1/4}\big] \\ &- P\big[\hat{\Delta}^{(3)}_{n,m} > c_{11}\{B^2_{n,m}\log(pm)\log^2(1/\alpha_{n,m})/m\}^{1/2}\big] \\ &- P\big[\hat{\Delta}^{(4)}_{n,m} > c_{20}\{B^2_{n,m}\log(pm)\log^2(1/\alpha_{n,m})/m\}^{1/4}\big] \geq 1 - \gamma_{n,m}, \end{aligned} \tag{54}$$

where $\gamma_{n,m} = (\alpha_{n,m})^{\log(pn)/3} + 3(\alpha_{n,m})^{\log^{1/2}(pn)/c_*} + (\alpha_{n,m})^{\log(pm)/3} + 3(\alpha_{n,m})^{\log^{1/2}(pm)/c_*} + (\alpha_{n,m})^{\log^3(pn)/6} + 3(\alpha_{n,m})^{\log^3(pn)/c_*} + (\alpha_{n,m})^{\log^3(pm)/6} + 3(\alpha_{n,m})^{\log^3(pm)/c_*}$ with $c_* = \max\{c_9, c_{12}, c_{18}, c_{21}\}$, and the second inequality is by (a). Finally, based on (35), (53) and (54), it can be concluded that with probability at least $1 - \gamma_{n,m}$, we have $\rho^{MB}_{n,m} \lesssim \{B^2_{n,m}\log^5(pn)\log^2(1/\alpha_{n,m})/n\}^{1/6} + \{B^2_{n,m}\log^5(pm)\log^2(1/\alpha_{n,m})/m\}^{1/6}$, which completes the proof. □

Proof of Corollary 1: First, based on Theorem 2, we know that with probability at least $1 - \gamma_{n,m}$, one has $\rho^{MB}_{n,m} \lesssim \{B^2_{n,m}\log^5(pn)\log^2(1/\alpha_{n,m})/n\}^{1/6} + \{B^2_{n,m}\log^5(pm)\log^2(1/\alpha_{n,m})/m\}^{1/6}$. Accordingly, we let $\{A_{n,m} : n \geq 1, m \geq 1, m \in \sigma(n)\}$ be a sequence of events where $A_{n,m}$ is the event such that $\rho^{MB}_{n,m} \lesssim \{B^2_{n,m}\log^5(pn)\log^2(1/\alpha_{n,m})/n\}^{1/6} + \{B^2_{n,m}\log^5(pm)\log^2(1/\alpha_{n,m})/m\}^{1/6}$. Note that $\sigma(n)$ is a set of positive integers determined by n and the conditions in

the Theorem, possibly the empty set. It then follows that $P(A_{n,m}) \geq 1-\gamma_{n,m}$, which further implies that $P(A_{n,m}^c) \leq \gamma_{n,m}$. Together with Lemma 6 and condition (f), it can be concluded that $P\big(\bigcup_{k_1=1}^{\infty}\bigcup_{k_2=1}^{\infty}\bigcap_{n=k_1}^{\infty}\bigcap_{m\in\varrho(k_2)\cap\sigma(n)} A_{n,m}\big) = 1 - P\big(\bigcap_{k_1=1}^{\infty}\bigcap_{k_2=1}^{\infty}\bigcup_{n=k_1}^{\infty}\bigcup_{m\in\varrho(k_2)\cap\sigma(n)} A_{n,m}^c\big) = 1$, which completes the proof. □

3. Proofs of Lemmas. *Proof of Lemma 1*: It follows directly from Lemmas A.5 and A.6 in [1]. □

Proof of Lemma 2: Given $\phi_1, \phi_2 \geq 1$, we denote $\phi = \min\{\phi_1, \phi_2\}$. Moreover, we denote $\beta = \phi \log p$, and given any $y \in \mathbb{R}^p$, we define the function $F_\beta(w)$ according to (2) that depends implicitly on y. Based on $F_\beta(w)$ and ϕ, we construct a function $\kappa(w) = \varphi_0(\phi F_\beta(w))$ upon (4). We also construct a function $h(w,t)$ as

$$h(w,t) = 1\big\{-\phi^{-1} - t/\beta < \max_{1\leq j\leq p}(w_j - y_j) \leq \phi^{-1} + t/\beta\big\},$$

for all $w \in \mathbb{R}^p$ and $t > 0$. Then we let $V^n = \{V_1, \ldots, V_n\}$ be an independent copy of F^n, while $W^m = \{W_1, \ldots, W_m\}$ is defined to be an independent copy of G^m. Given any $v \in [0,1]$, we denote the Slepian interpolant as

$$Z(t) = Z^*(t) + Z^{**}(t), t \in [0,1],$$

where

$$Z^*(t) = \sum_{i=1}^{n} Z_i^*(t), t \in [0,1],$$
$$Z^{**}(t) = \sum_{i'=1}^{m} Z_{i'}^{**}(t), t \in [0,1],$$

Note that for all $i = 1, \ldots, n$, $i' = 1, \ldots, m$, $t \in [0,1]$

$$\begin{aligned} Z_i^*(t) =& (Z_{i1}^*(t), \ldots, Z_{ip}^*(t))' = n^{-1/2}\big[t^{1/2}\{v^{1/2}(X_i - \mu^X) + \\ & (1-v)^{1/2}(F_i - \mu^X)\} + (1-t)^{1/2}(V_i - \mu^X)\big], \\ Z_{i'}^{**}(t) =& (Z_{i'1}^{**}(t), \ldots, Z_{i'p}^{**}(t))' = m^{-1/2}\delta_{n,m}\big[t^{1/2}\{v^{1/2}(Y_{i'} - \mu^Y) + \\ & (1-v)^{1/2}(G_{i'} - \mu^Y)\} + (1-t)^{1/2}(W_{i'} - \mu^Y)\big]. \end{aligned}$$

Then, we define the leave-one-out terms for $Z^*(t)$ and $Z^{**}(t)$ as

$$Z^{*(i)}(t) = Z^*(t) - Z_i^*(t), t \in [0,1], i = 1, \ldots, n$$

and

$$Z^{**(i)}(t) = Z^{**}(t) - Z_i^{**}(t), t \in [0,1], i = 1, \ldots, m$$

respectively. For all $i = 1, \ldots, n$, $i' = 1, \ldots, m$, $t \in [0, 1]$, we denote

$$\begin{aligned}
\dot{Z}_i^*(t) =&(\dot{Z}_{i1}^*(t), \ldots, \dot{Z}_{ip}^*(t))' = n^{-1/2}\big[t^{-1/2}\{v^{1/2}(X_i - \mu^X)+ \\
&(1-v)^{1/2}(F_i - \mu^X)\} - (1-t)^{-1/2}(V_i - \mu^X)\big], \\
\dot{Z}_{i'}^{**}(t) =&(\dot{Z}_{i'1}^{**}(t), \ldots, \dot{Z}_{i'p}^{**}(t))' = m^{-1/2}\delta_{n,m}\big[t^{-1/2}\{v^{1/2}(Y_{i'} - \mu^Y)+ \\
&(1-v)^{1/2}(G_{i'} - \mu^Y)\} - (1-t)^{-1/2}(W_{i'} - \mu^Y)\big].
\end{aligned}$$

Note that we sometimes omit the t and write $Z = Z(t)$, $\dot{Z}_i^* = \dot{Z}_i^*(t)$, etc, for the sake of brevity. It can be shown that

$$\begin{aligned}
Z(1) =&v^{1/2}(S_n^X - n^{1/2}\mu^X) + (1-v)^{1/2}(S_n^F - n^{1/2}\mu^X)+ \\
&v^{1/2}\delta_{n,m}(S_m^Y - m^{1/2}\mu^Y) + (1-v)^{1/2}\delta_{n,m}(S_m^G - m^{1/2}\mu^Y), \\
Z(0) =&S_n^V - n^{1/2}\mu^X + \delta_{n,m}S_m^W - \delta_{n,m}m^{1/2}\mu^Y.
\end{aligned}$$

We further denote a quantity $\mathcal{I}_{n,m}$ as

$$\begin{aligned}
\mathcal{I}_{n,m} =&\kappa(v^{1/2}(S_n^X - n^{1/2}\mu^X) + (1-v)^{1/2}(S_n^F - n^{1/2}\mu^X)+ \\
&v^{1/2}\delta_{n,m}(S_m^Y - m^{1/2}\mu^Y) + (1-v)^{1/2}\delta_{n,m}(S_m^G - m^{1/2}\mu^Y))- \\
&\kappa(S_n^V - n^{1/2}\mu^X + \delta_{n,m}S_m^W - \delta_{n,m}m^{1/2}\mu^Y) \\
=&\kappa(Z(1)) - \kappa(Z(0))
\end{aligned}$$

Finally, we denote λ_i^*, λ_i^{**} as

$$\begin{aligned}
\lambda_i^* =&1\Big\{\max_{1\le j\le p}|X_{ij} - \mu_j^X| \vee |F_{ij} - \mu_j^X| \vee |V_{ij} - \mu_j^X| \le \\
&n^{1/2}/(4\beta)\Big\}, \quad i = 1, \ldots, n, \\
\lambda_i^{**} =&1\Big\{\max_{1\le j\le p}|Y_{ij} - \mu_j^Y| \vee |G_{ij} - \mu_j^Y| \vee |W_{ij} - \mu_j^Y| \le \\
&m^{1/2}/(4\beta|\delta_{n,m}|)\Big\}, \quad i = 1, \ldots, m.
\end{aligned}$$

Now, we are in a position to start the proof. According to the above notations, it can be seen that

$$\begin{aligned}
\mathcal{I}_{n,m} &= \kappa(Z(1)) - \kappa(Z(0)) = \int_0^1 \frac{d\kappa(Z(t))}{dt}dt \\
&= 2^{-1}\sum_{j=1}^p\sum_{i=1}^n\int_0^1 \kappa_j(Z)\dot{Z}_{ij}^* dt + 2^{-1}\sum_{j'=1}^p\sum_{i'=1}^m\int_0^1 \kappa_{j'}(Z)\dot{Z}_{i'j'}^{**}dt \\
&= \mathcal{I}_{n,m}^* + \mathcal{I}_{n,m}^{**}, \qquad (55)
\end{aligned}$$

where

$$\mathcal{I}^*_{n,m} = 2^{-1}\sum_{j=1}^{p}\sum_{i=1}^{n}\int_0^1 \kappa_j(Z)\dot{Z}^*_{ij}dt \tag{56}$$

and

$$\mathcal{I}^{**}_{n,m} = 2^{-1}\sum_{j=1}^{p}\sum_{i=1}^{m}\int_0^1 \kappa_j(Z)\dot{Z}^{**}_{ij}dt. \tag{57}$$

Moreover, by Taylor's theorem, it can be seen that for all $j = 1, \ldots, p$ and for any $t \in [0, 1]$

$$\begin{aligned}\kappa_j(Z) = \kappa_j(Z^{*(i)} + Z^{**}) + \sum_{k=1}^{p}\kappa_{jk}(Z^{*(i)} + Z^{**})Z^*_{ik} + \\ \sum_{k=1}^{p}\sum_{l=1}^{p}\int_0^1 (1-\tau)\kappa_{jkl}(Z^{*(i)} + Z^{**} + \tau Z^*_i)Z^*_{ik}Z^*_{il}d\tau, \quad i = 1, \ldots, n\end{aligned} \tag{58}$$

and

$$\begin{aligned}\kappa_j(Z) = \kappa_j(Z^{**(i)} + Z^{*}) + \sum_{k=1}^{p}\kappa_{jk}(Z^{**(i)} + Z^{*})Z^{**}_{ik} + \\ \sum_{k=1}^{p}\sum_{l=1}^{p}\int_0^1 (1-\tau)\kappa_{jkl}(Z^{**(i)} + Z^{*} + \tau Z^{**}_i)Z^{**}_{ik}Z^{**}_{il}d\tau, \quad i = 1, \ldots, m.\end{aligned} \tag{59}$$

By combining (56) with (58), it can be seen that

$$\begin{aligned}\mathcal{I}^*_{n,m} =& 2^{-1}\sum_{j=1}^{p}\sum_{i=1}^{n}\int_0^1 \kappa_j(Z^{*(i)} + Z^{**})\dot{Z}^*_{ij}dt \\ &+ 2^{-1}\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{i=1}^{n}\int_0^1 \kappa_{jk}(Z^{*(i)} + Z^{**})\dot{Z}^*_{ij}Z^*_{ik}dt \\ &+ 2^{-1}\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 (1-\tau)\kappa_{jkl}(Z^{*(i)} + Z^{**} + \tau Z^*_i) \\ &\cdot \dot{Z}^*_{ij}Z^*_{ik}Z^*_{il}d\tau dt,\end{aligned}$$

which implies that

$$E(\mathcal{I}^*_{n,m}) = 2^{-1}(I^* + II^* + III^*), \tag{60}$$

where

$$
\begin{aligned}
I^* &= \sum_{j=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\kappa_j(Z^{*(i)}+Z^{**})\dot{Z}^*_{ij}\}dt,\\
II^* &= \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\kappa_{jk}(Z^{*(i)}+Z^{**})\dot{Z}^*_{ij}Z^*_{ik}\}dt,\\
III^* &= \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 (1-\tau)E\{\kappa_{jkl}(Z^{*(i)}+Z^{**}+\tau Z^*_i)\\
&\quad \cdot \dot{Z}^*_{ij}Z^*_{ik}Z^*_{il}\}d\tau dt.
\end{aligned}
$$

For I^*, we have

$$
\begin{aligned}
I^* &= \sum_{j=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\kappa_j(Z^{*(i)}+Z^{**})\dot{Z}^*_{ij}\}dt\\
&= \sum_{j=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\kappa_j(Z^{*(i)}+Z^{**})\}E(\dot{Z}^*_{ij})dt\\
&= 0, \qquad (61)
\end{aligned}
$$

where the second equality follows from the independence of $Z^{*(i)}+Z^{**}$ from $\dot{Z}^*_{ij}$, and the last equality is by $E(\dot{Z}^*_{ij})=0$. For II^*, we have

$$
\begin{aligned}
II^* &= \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\kappa_{jk}(Z^{*(i)}+Z^{**})\dot{Z}^*_{ij}Z^*_{ik}\}dt\\
&= \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\kappa_{jk}(Z^{*(i)}+Z^{**})\}E(\dot{Z}^*_{ij}Z^*_{ik})dt, \qquad (62)
\end{aligned}
$$

where the last equality follows from the independence of $Z^{*(i)}+Z^{**}$ from $\dot{Z}^*_{ij}Z^*_{ik}$.

Moreover, for every $i = 1, \ldots, n$; $j, k = 1, \ldots, p$ and $t \in [0, 1]$, we have

$$\begin{aligned}
E(\dot{Z}^*_{ij} Z^*_{ik}) =& E\Big(n^{-1}\big[t^{-1/2}\{v^{1/2}(X_{ij} - \mu^X_j) + (1-v)^{1/2}(F_{ij} - \mu^X_j)\} - \\
& (1-t)^{-1/2}(V_{ij} - \mu^X_j)\big] \cdot \big[t^{1/2}\{v^{1/2}(X_{ik} - \mu^X_k) + \\
& (1-v)^{1/2}(F_{ik} - \mu^X_k)\} + (1-t)^{1/2}(V_{ik} - \mu^X_k)\big]\Big) \\
=& n^{-1} E\Big[\{v^{1/2}(X_{ij} - \mu^X_j) + (1-v)^{1/2}(F_{ij} - \mu^X_j)\} \\
& \cdot \{v^{1/2}(X_{ik} - \mu^X_k) + (1-v)^{1/2}(F_{ik} - \mu^X_k)\} - \\
& E\{(V_{ij} - \mu^X_j)(V_{ik} - \mu^X_k)\}\Big] \\
=& n^{-1} E\{v(X_{ij} - \mu^X_j)(X_{ik} - \mu^X_k) + \\
& (1-v)(F_{ij} - \mu^X_j)(F_{ik} - \mu^X_k) - (V_{ij} - \mu^X_j)(V_{ik} - \mu^X_k)\} \\
=& 0, \qquad (63)
\end{aligned}$$

where the second and the third equalities follow from the independence among X^n, F^n, V^n, and the last equality holds since $E\{(X_{ij} - \mu^X_j)(X_{ik} - \mu^X_k)\} = E\{(F_{ij} - \mu^X_j)(F_{ik} - \mu^X_k)\} = E\{(V_{ij} - \mu^X_j)(V_{ik} - \mu^X_k)\}$. Hence, by combining (62) with (63), we conclude that

$$II^* = 0. \qquad (64)$$

For III^*, we can write it as

$$III^* = III^*_1 + III^*_2, \qquad (65)$$

where

$$\begin{aligned}
III^*_1 =& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 (1-\tau) E\{\lambda^*_i \kappa_{jkl}(Z^{*(i)} + Z^{**} + \tau Z^*_i) \\
& \cdot \dot{Z}^*_{ij} Z^*_{ik} Z^*_{il}\} d\tau dt, \\
III^*_2 =& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 (1-\tau) E\{(1-\lambda^*_i) \kappa_{jkl}(Z^{*(i)} + Z^{**} + \tau Z^*_i) \\
& \cdot \dot{Z}^*_{ij} Z^*_{ik} Z^*_{il}\} d\tau dt.
\end{aligned}$$

For III_2^*, we have

$$
\begin{aligned}
|III_2^*| \leq & \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 (1-\tau)E\{(1-\lambda_i^*)|\kappa_{jkl}(Z^{*(i)}+Z^{**}+\tau Z_i^*)| \\
& \cdot |\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|\}d\tau dt \\
\leq & \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 (1-\tau)E\{(1-\lambda_i^*)Q_{jkl}(Z^{*(i)}+Z^{**}+\tau Z_i^*) \\
& \cdot |\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|\}d\tau dt \\
\lesssim & \phi\beta^2 \sum_{i=1}^{n}\int_0^1 E\big\{\big(1-\lambda_i^*\big)\big(\max_{1\leq j,k,l\leq p}|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|\big)\big\}dt, \qquad (66)
\end{aligned}
$$

where the first inequality follows from Jensen's inequality, and the second and the third inequalities follow from Lemma 1. Moreover, note that for all $i = 1, \ldots, n$, we have

$$
\begin{aligned}
1-\lambda_i^* = & 1\big\{\max_{1\leq j\leq p}|X_{ij}-\mu_j^X| \vee |F_{ij}-\mu_j^X| \vee |V_{ij}-\mu_j^X| > n^{1/2}/(4\beta)\big\} \\
\leq & 1\big\{\max_{1\leq j\leq p}|X_{ij}-\mu_j^X| > n^{1/2}/(4\beta)\big\} + \\
& 1\big\{\max_{1\leq j\leq p}|F_{ij}-\mu_j^X| > n^{1/2}/(4\beta)\big\} + \\
& 1\big\{\max_{1\leq j\leq p}|V_{ij}-\mu_j^X| > n^{1/2}/(4\beta)\big\}. \qquad (67)
\end{aligned}
$$

Also note that for all $i = 1, \ldots, n$ and for any $t \in [0, 1]$, we have

$$
\begin{aligned}
&\max_{1\le j,k,l\le p} |\dot{Z}^*_{ij} Z^*_{ik} Z^*_{il}| \\
=&\max_{1\le j,k,l\le p} \Big| n^{-1/2}\big[t^{-1/2}\{v^{1/2}(X_{ij}-\mu_j^X)+(1-v)^{1/2}(F_{ij}-\mu_j^X)\}- \\
&(1-t)^{-1/2}(V_{ij}-\mu_j^X)\big]\Big| \cdot \Big| n^{-1/2}\big[t^{1/2}\{v^{1/2}(X_{ik}-\mu_k^X)+ \\
&(1-v)^{1/2}(F_{ik}-\mu_k^X)\}+(1-t)^{1/2}(V_{ik}-\mu_k^X)\big]\Big| \\
&\cdot \Big| n^{-1/2}\big[t^{1/2}\{v^{1/2}(X_{il}-\mu_l^X)+(1-v)^{1/2}(F_{il}-\mu_l^X)\}+ \\
&(1-t)^{1/2}(V_{il}-\mu_l^X)\big]\Big| \\
\le& 100 n^{-3/2}\{t^{1/2}\wedge(1-t)^{1/2}\}^{-1} \\
&\cdot \max_{1\le j\le p} |X_{ij}-\mu_j^X|^3 \vee |F_{ij}-\mu_j^X|^3 \vee |V_{ij}-\mu_j^X|^3 \\
\le& 100 n^{-3/2}\{t^{1/2}\wedge(1-t)^{1/2}\}^{-1} \\
&\cdot \big(\max_{1\le j\le p} |X_{ij}-\mu_j^X|^3 + \max_{1\le j\le p} |F_{ij}-\mu_j^X|^3 + \max_{1\le j\le p} |V_{ij}-\mu_j^X|^3\big). \qquad (68)
\end{aligned}
$$

Based on (66), (67) and (68), it can be deduced that

$$
\begin{aligned}
|III_2^*| \lesssim & n^{-3/2}\phi\beta^2\Big[\int_0^1 \{t^{1/2}\wedge(1-t)^{1/2}\}^{-1}dt\Big]\cdot \\
& \sum_{i=1}^n E\Big(\Big[1\big\{\max_{1\le j\le p}|X_{ij}-\mu_j^X|>n^{1/2}/(4\beta)\big\}+ \\
& 1\big\{\max_{1\le j\le p}|F_{ij}-\mu_j^X|>n^{1/2}/(4\beta)\big\}+ \\
& 1\big\{\max_{1\le j\le p}|V_{ij}-\mu_j^X|>n^{1/2}/(4\beta)\big\}\Big] \\
& \cdot\Big[\max_{1\le j\le p}|X_{ij}-\mu_j^X|^3+\max_{1\le j\le p}|F_{ij}-\mu_j^X|^3+\max_{1\le j\le p}|V_{ij}-\mu_j^X|^3\Big]\Big) \\
\lesssim & n^{-3/2}\phi\beta^2\sum_{i=1}^n E\Big(\Big[1\big\{\max_{1\le j\le p}|X_{ij}-\mu_j^X|>n^{1/2}/(4\beta)\big\}+ \\
& 1\big\{\max_{1\le j\le p}|F_{ij}-\mu_j^X|>n^{1/2}/(4\beta)\big\}+ \\
& 1\big\{\max_{1\le j\le p}|V_{ij}-\mu_j^X|>n^{1/2}/(4\beta)\big\}\Big] \\
& \cdot\Big[\max_{1\le j\le p}|X_{ij}-\mu_j^X|^3+\max_{1\le j\le p}|F_{ij}-\mu_j^X|^3+\max_{1\le j\le p}|V_{ij}-\mu_j^X|^3\Big]\Big) \\
\lesssim & n^{-1/2}\phi\beta^2\big\{M_n^X(\phi)+M_n^F(\phi)+M_n^V(\phi)\big\} \\
\lesssim & n^{-1/2}\phi\beta^2\big\{M_n^X(\phi)+M_n^F(\phi)\big\} \\
\lesssim & n^{-1/2}\phi\beta^2 M_n(\phi), \qquad (69)
\end{aligned}
$$

where the third inequality is by Lemma B.1 in [3], and the fourth inequality follows

from the fact that $M_n^F(\phi) = M_n^V(\phi)$. For III_1^*, we have

$$\begin{aligned}
|III_1^*| \leq & \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 (1-\tau)E\{\lambda_i^*|\kappa_{jkl}(Z^{*(i)}+Z^{**}+\tau Z_i^*)| \\
& \cdot |\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|\}d\tau dt \\
\leq & \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 E\{\lambda_i^*|\kappa_{jkl}(Z^{*(i)}+Z^{**}+\tau Z_i^*)| \\
& \cdot |\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|\}d\tau dt \\
= & \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 E\{\lambda_i^* h(Z^{*(i)}+Z^{**},1) \\
& \cdot |\kappa_{jkl}(Z^{*(i)}+Z^{**}+\tau Z_i^*)| \cdot |\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|\}d\tau dt \\
\lesssim & \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1\int_0^1 E\{\lambda_i^* h(Z^{*(i)}+Z^{**},1)Q_{jkl}(Z^{*(i)}+Z^{**}) \\
& \cdot |\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|\}d\tau dt \\
\lesssim & \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{h(Z^{*(i)}+Z^{**},1)Q_{jkl}(Z^{*(i)}+Z^{**}) \\
& \cdot |\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|\}dt \\
\lesssim & \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{h(Z^{*(i)}+Z^{**},1)Q_{jkl}(Z^{*(i)}+Z^{**})\} \\
& \cdot E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)dt \\
\lesssim & \Delta_1^* + \Delta_2^*, \qquad (70)
\end{aligned}$$

where the equality follows from the fact that

$$\begin{aligned}
& \lambda_i^* h(Z^{*(i)}+Z^{**},1)|\kappa_{jkl}(Z^{*(i)}+Z^{**}+\tau Z_i^*)| \\
= & \lambda_i^*|\kappa_{jkl}(Z^{*(i)}+Z^{**}+\tau Z_i^*)|, \quad i=1,\ldots,n
\end{aligned}$$

whenever $\tau \in [0,1]$, and the third inequality is by Lemma 1. In addition, the terms

Δ_1^* and Δ_2^* are given by

$$\begin{aligned}\Delta_1^* =& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{(1-\lambda_i^*)h(Z^{*(i)}+Z^{**},1)Q_{jkl}(Z^{*(i)}+Z^{**})\}\\ &\cdot E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)dt,\\ \Delta_2^* =& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\lambda_i^* h(Z^{*(i)}+Z^{**},1)Q_{jkl}(Z^{*(i)}+Z^{**})\}\\ &\cdot E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)dt.\end{aligned}$$

For Δ_1^*, we have

$$\begin{aligned}\Delta_1^* =& \sum_{i=1}^{n}\int_0^1 \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p} E\{(1-\lambda_i^*)h(Z^{*(i)}+Z^{**},1)Q_{jkl}(Z^{*(i)}+Z^{**})\}\\ &\cdot E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)dt\\ \le& \sum_{i=1}^{n}\int_0^1 \Big\{\max_{1\le j,k,l\le p} E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)\Big\}\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p} E\big\{(1-\lambda_i^*)\\ &\cdot h(Z^{*(i)}+Z^{**},1)Q_{jkl}(Z^{*(i)}+Z^{**})\big\}dt\\ \lesssim& \phi\beta^2\sum_{i=1}^{n}\int_0^1 \Big\{\max_{1\le j,k,l\le p} E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)\Big\}E\big\{(1-\lambda_i^*)\\ &\cdot h(Z^{*(i)}+Z^{**},1)\big\}dt\\ \lesssim& \phi\beta^2\sum_{i=1}^{n}\int_0^1 \Big\{\max_{1\le j,k,l\le p} E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)\Big\}\big\{E(1-\lambda_i^*)\big\}dt\\ \lesssim& n^{-1/2}\phi\beta^2\Big[\int_0^1 \{t^{1/2}\wedge(1-t)^{1/2}\}^{-1}dt\Big]\\ &\cdot\big\{M_n^X(\phi)+M_n^F(\phi)+M_n^V(\phi)\big\}\\ \lesssim& n^{-1/2}\phi\beta^2\big\{M_n^X(\phi)+M_n^F(\phi)\big\}\\ \lesssim& n^{-1/2}\phi\beta^2 M_n(\phi),\end{aligned} \tag{71}$$

where the first inequality is by Holder's inequality, the second inequality is due to Lemma 1, and the fourth inequality is based on (67), (68) and Lemma B.1 in [3].

For Δ_2^*, we have

$$\begin{aligned}
\Delta_2^* =& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\lambda_i^* h(Z^{*(i)}+Z^{**},1)Q_{jkl}(Z^{*(i)}+Z^{**})\} \\
&\cdot E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)dt \\
\lesssim& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\lambda_i^* h(Z^{*(i)}+Z^{**},1)Q_{jkl}(Z^{*}+Z^{**})\} \\
&\cdot E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)dt \\
\lesssim& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{\lambda_i^* h(Z^{*(i)}+Z^{**},1)h(Z^{*}+Z^{**},2) \\
&\cdot Q_{jkl}(Z^{*}+Z^{**})\}E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)dt \\
\lesssim& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{n}\int_0^1 E\{h(Z^{*}+Z^{**},2)Q_{jkl}(Z^{*}+Z^{**})\} \\
&\cdot E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)dt \\
\lesssim& \phi\beta^2 \int_0^1 E\{h(Z^{*}+Z^{**},2)\}\big\{\max_{1\le j,k,l\le p}\sum_{i=1}^{n} E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)\big\}dt, \qquad (72)
\end{aligned}$$

where the first inequality is by Lemma 1, the second inequality follows from the fact that

$$\lambda_i^* h(Z^{*(i)}+Z^{**},1) = \lambda_i^* h(Z^{*(i)}+Z^{**},1)h(Z^{*}+Z^{**},2), \quad i=1,\ldots,n,$$

and the fourth inequality is based on Holder's inequality and Lemma 1. Furthermore,

$$\begin{aligned}
&\max_{1\le j,k,l\le p}\sum_{i=1}^{n} E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|) \\
\lesssim& n^{-3/2}\{t^{1/2}\wedge(1-t)^{1/2}\}^{-1} \\
&\cdot \max_{1\le j\le p}\sum_{i=1}^{n} E(|X_{ij}-\mu_j^X|^3+|F_{ij}-\mu_j^X|^3+|V_{ij}-\mu_j^X|^3) \\
\lesssim& n^{-3/2}\{t^{1/2}\wedge(1-t)^{1/2}\}^{-1} \\
&\cdot \big\{\max_{1\le j\le p}\sum_{i=1}^{n} E(|X_{ij}-\mu_j^X|^3)+\max_{1\le j\le p}\sum_{i=1}^{n} E(|F_{ij}-\mu_j^X|^3)\big\}.
\end{aligned}$$

Together with the fact that $E(|F_{ij}-\mu_j^X|^3) \lesssim \{E(|F_{ij}-\mu_j^X|^2)\}^{3/2} \lesssim \{E(|X_{ij}-\mu_j^X|^2)\}^{3/2} \lesssim E(|X_{ij}-\mu_j^X|^3)$, we conclude that

$$\max_{1\le j,k,l\le p}\sum_{i=1}^{n} E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|) \lesssim n^{-1/2}\{t^{1/2}\wedge(1-t)^{1/2}\}^{-1} L_n^X. \tag{73}$$

Moreover, by definition of the function h, we have

$$\begin{aligned} E\{h(Z^*+Z^{**},2)\} =& P(Z^*+Z^{**}-y\le \phi^{-1}+2\beta^{-1})- \\ & P(Z^*+Z^{**}-y\le -\phi^{-1}-2\beta^{-1}). \end{aligned} \tag{74}$$

In addition, it can be verified that

$$\begin{aligned} Z^*+Z^{**} \overset{\mathrm{d}}{=}& (tv)^{1/2}(S_n^X+\delta_{n,m}S_m^Y-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y)+ \\ & (1-tv)^{1/2}(S_n^F+\delta_{n,m}S_m^G-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y), \end{aligned} \tag{75}$$

where $\overset{\mathrm{d}}{=}$ means having the same distribution. Therefore, under (75), we have

$$\begin{aligned} & P(Z^*+Z^{**}-y\le \phi^{-1}+2\beta^{-1}) \\ =& P\big\{(tv)^{1/2}(S_n^X+\delta_{n,m}S_m^Y-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y)+ \\ & (1-tv)^{1/2}(S_n^F+\delta_{n,m}S_m^G-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y)-y\le \\ & \phi^{-1}+2\beta^{-1}\big\} \\ =& P\big\{(tv)^{1/2}(S_n^X+\delta_{n,m}S_m^Y-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y)+ \\ & (1-tv)^{1/2}(S_n^F+\delta_{n,m}S_m^G-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y)\le \\ & y+\phi^{-1}+2\beta^{-1}\big\} \\ =& P\big\{(tv)^{1/2}(S_n^X+\delta_{n,m}S_m^Y-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y)+ \\ & (1-tv)^{1/2}(S_n^F+\delta_{n,m}S_m^G-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y)\le \\ & y+\phi^{-1}+2\beta^{-1}\big\} \\ & -P(S_n^F+\delta_{n,m}S_m^G-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y\le \\ & y+\phi^{-1}+2\beta^{-1}) \\ & +P(S_n^F+\delta_{n,m}S_m^G-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y\le \\ & y+\phi^{-1}+2\beta^{-1}) \\ \le& \rho_{n,m}+P(S_n^F+\delta_{n,m}S_m^G-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y\le \\ & y+\phi^{-1}+2\beta^{-1}). \end{aligned} \tag{76}$$

Likewise, we have

$$
\begin{aligned}
&P(Z^* + Z^{**} - y \le -\phi^{-1} - 2\beta^{-1}) \\
=&P\big\{(tv)^{1/2}(S_n^X + \delta_{n,m}S_m^Y - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y) + \\
&(1-tv)^{1/2}(S_n^F + \delta_{n,m}S_m^G - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y) - y \le \\
&-\phi^{-1} - 2\beta^{-1}\big\} \\
=&P\big\{(tv)^{1/2}(S_n^X + \delta_{n,m}S_m^Y - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y) + \\
&(1-tv)^{1/2}(S_n^F + \delta_{n,m}S_m^G - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y) \le \\
&y - \phi^{-1} - 2\beta^{-1}\big\} \\
=&P\big\{(tv)^{1/2}(S_n^X + \delta_{n,m}S_m^Y - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y) + \\
&(1-tv)^{1/2}(S_n^F + \delta_{n,m}S_m^G - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y) \le \\
&y - \phi^{-1} - 2\beta^{-1}\big\} \\
&- P(S_n^F + \delta_{n,m}S_m^G - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y \le \\
&y - \phi^{-1} - 2\beta^{-1}) \\
&+ P(S_n^F + \delta_{n,m}S_m^G - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y \le \\
&y - \phi^{-1} - 2\beta^{-1}) \\
\ge& -\rho_{n,m} + P(S_n^F + \delta_{n,m}S_m^G - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y \le \\
&y - \phi^{-1} - 2\beta^{-1}). \qquad (77)
\end{aligned}
$$

Based on (74), (76) and (77), we have

$$
\begin{aligned}
&E\{h(Z^* + Z^{**}, 2)\} \\
\le& 2\rho_{n,m} + P(S_n^F + \delta_{n,m}S_m^G - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y \le \\
&y + \phi^{-1} + 2\beta^{-1}) \\
&- P(S_n^F + \delta_{n,m}S_m^G - n^{1/2}\mu^X - \delta_{n,m}m^{1/2}\mu^Y \le \\
&y - \phi^{-1} - 2\beta^{-1}) \\
\lesssim& \rho_{n,m} + (\phi^{-1} + 2\beta^{-1})(\log p)^{1/2} \\
\lesssim& \rho_{n,m} + \phi^{-1}(\log p)^{1/2}, \qquad (78)
\end{aligned}
$$

where the second inequality follows from combining condition (a) with Lemma A.1 in [3]. Therefore, based on (72), (73) and (78), it can be verified that

$$
\begin{aligned}
\Delta_2^* &\lesssim \phi\beta^2 \int_0^1 E\{h(Z^* + Z^{**}, 2)\}\big\{\max_{1\le j,k,l\le p}\sum_{i=1}^n E(|\dot{Z}_{ij}^* Z_{ik}^* Z_{il}^*|)\big\}dt \\
&\lesssim n^{-1/2}\phi\beta^2\big\{L_n^X\rho_{n,m} + \phi^{-1}L_n^X(\log p)^{1/2}\big\}. \qquad (79)
\end{aligned}
$$

Under (65), (69), (70), (71) and (79), it can be deduced that

$$
\begin{aligned}
|III^*| &\lesssim n^{-1/2}\phi\beta^2 M_n(\phi) + n^{-1/2}\phi\beta^2\big\{L_n^X\rho_{n,m} + \phi^{-1}L_n^X(\log p)^{1/2}\big\} \\
&\lesssim n^{-1/2}\beta^2\big\{\phi L_n^X\rho_{n,m} + L_n^X(\log p)^{1/2} + \phi M_n(\phi)\big\} \\
&\lesssim n^{-1/2}\phi^2(\log p)^2\big\{\phi L_n^X\rho_{n,m} + L_n^X(\log p)^{1/2} + \phi M_n(\phi)\big\} \\
&\lesssim n^{-1/2}\phi_1^2(\log p)^2\big\{\phi_1 L_n^X\rho_{n,m} + L_n^X(\log p)^{1/2} + \phi_1 M_n(\phi_1)\big\}, \qquad (80)
\end{aligned}
$$

where the last inequality follows from the facts that $\phi_1 \geq \phi$ and $M_n(\phi_1) \geq M_n(\phi)$. Hence by combining (60), (61), (64) with (80), it is clear that

$$
\begin{aligned}
|E(\mathcal{I}^*_{n,m})| \leq& 2^{-1}|III^*| \\
\lesssim& n^{-1/2}\phi_1^2(\log p)^2\big\{\phi_1 L_n^X\rho_{n,m} + L_n^X(\log p)^{1/2} + \phi_1 M_n(\phi_1)\big\}. \qquad (81)
\end{aligned}
$$

Next, we start to bound the quantity $|E(\mathcal{I}^{**}_{n,m})|$. Similarly, by combining (57) with (59), we have

$$
\begin{aligned}
\mathcal{I}^{**}_{n,m} =& 2^{-1}\sum_{j=1}^{p}\sum_{i=1}^{m}\int_0^1 \kappa_j(Z^{**(i)} + Z^*)\dot{Z}^{**}_{ij}dt + \\
& 2^{-1}\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{i=1}^{m}\int_0^1 \kappa_{jk}(Z^{**(i)} + Z^*)\dot{Z}^{**}_{ij}Z^{**}_{ik}dt + \\
& 2^{-1}\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 (1-\tau)\kappa_{jkl}(Z^{**(i)} + Z^* + \tau Z^{**}_i) \\
& \cdot \dot{Z}^{**}_{ij}Z^{**}_{ik}Z^{**}_{il}d\tau dt,
\end{aligned}
$$

which implies that

$$
E(\mathcal{I}^{**}_{n,m}) = 2^{-1}(I^{**} + II^{**} + III^{**}), \qquad (82)
$$

where

$$
\begin{aligned}
I^{**} =& \sum_{j=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\kappa_j(Z^{**(i)} + Z^*)\dot{Z}^{**}_{ij}\}dt, \\
II^{**} =& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\kappa_{jk}(Z^{**(i)} + Z^*)\dot{Z}^{**}_{ij}Z^{**}_{ik}\}dt, \\
III^{**} =& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 (1-\tau)E\{\kappa_{jkl}(Z^{**(i)} + Z^* + \tau Z^{**}_i) \\
& \cdot \dot{Z}^{**}_{ij}Z^{**}_{ik}Z^{**}_{il}\}d\tau dt.
\end{aligned}
$$

For I^{**}, we have

$$\begin{aligned} I^{**} &= \sum_{j=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\kappa_j(Z^{**(i)}+Z^*)\dot{Z}_{ij}^{**}\}dt \\ &= \sum_{j=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\kappa_j(Z^{**(i)}+Z^*)\}E(\dot{Z}_{ij}^{**})dt \\ &= 0, \end{aligned} \tag{83}$$

where the second equality follows from the independence of $Z^{**(i)}+Z^*$ from $\dot{Z}_{ij}^{**}$, and the last equality is by $E(\dot{Z}_{ij}^{**})=0$. For II^{**}, we have

$$\begin{aligned} II^{**} &= \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\kappa_{jk}(Z^{**(i)}+Z^*)\dot{Z}_{ij}^{**}Z_{ik}^{**}\}dt \\ &= \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\kappa_{jk}(Z^{**(i)}+Z^*)\}E(\dot{Z}_{ij}^{**}Z_{ik}^{**})dt, \end{aligned} \tag{84}$$

where the last equality is based on the independence of $Z^{**(i)}+Z^*$ from $\dot{Z}_{ij}^{**}Z_{ik}^{**}$. In addition, for all $i=1,\ldots,m$, $j,k=1,\ldots,p$, we have

$$\begin{aligned} E(\dot{Z}_{ij}^{**}Z_{ik}^{**}) =& m^{-1}\delta_{n,m}^2 E\Big(\big[t^{-1/2}\{v^{1/2}(Y_{ij}-\mu_j^Y)+(1-v)^{1/2}(G_{ij}-\mu_j^Y)\}- \\ & (1-t)^{-1/2}(W_{ij}-\mu_j^Y)\big]\cdot\big[t^{1/2}\{v^{1/2}(Y_{ik}-\mu_k^Y)+ \\ & (1-v)^{1/2}(G_{ik}-\mu_k^Y)\}+(1-t)^{1/2}(W_{ik}-\mu_k^Y)\big]\Big) \\ =& m^{-1}\delta_{n,m}^2 E\Big[\{v^{1/2}(Y_{ij}-\mu_j^Y)+(1-v)^{1/2}(G_{ij}-\mu_j^Y)\} \\ & \cdot\{v^{1/2}(Y_{ik}-\mu_k^Y)+(1-v)^{1/2}(G_{ik}-\mu_k^Y)\}- \\ & (W_{ij}-\mu_j^Y)(W_{ik}-\mu_k^Y)\Big] \\ =& m^{-1}\delta_{n,m}^2 E\{v(Y_{ij}-\mu_j^Y)(Y_{ik}-\mu_k^Y)+ \\ & (1-v)(G_{ij}-\mu_j^Y)(G_{ik}-\mu_k^Y)-(W_{ij}-\mu_j^Y)(W_{ik}-\mu_k^Y)\} \\ =& 0, \end{aligned} \tag{85}$$

where the second and the third equalities are based on the independence among Y^m, G^m, W^m, and the last equality follows from $E\{(Y_{ij}-\mu_j^Y)(Y_{ik}-\mu_k^Y)\}=E\{(G_{ij}-\mu_j^Y)(G_{ik}-\mu_k^Y)\}=E\{(W_{ij}-\mu_j^Y)(W_{ik}-\mu_k^Y)\}$. Thus, by combining (84) with (85), we have

$$II^{**}=0. \tag{86}$$

For III^{**}, it could be decomposed as

$$III^{**} = III_1^{**} + III_2^{**}, \tag{87}$$

where

$$\begin{aligned} III_1^{**} =& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 (1-\tau)E\{\lambda_i^{**}\kappa_{jkl}(Z^{**(i)} + Z^* + \tau Z_i^{**}) \\ & \cdot \dot{Z}_{ij}^{**} Z_{ik}^{**} Z_{il}^{**}\}d\tau dt, \\ III_2^{**} =& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 (1-\tau)E\{(1-\lambda_i^{**})\kappa_{jkl}(Z^{**(i)} + Z^* + \tau Z_i^{**}) \\ & \cdot \dot{Z}_{ij}^{**} Z_{ik}^{**} Z_{il}^{**}\}d\tau dt. \end{aligned}$$

For III_2^{**}, it is easy to observe that

$$\begin{aligned} |III_2^{**}| \le& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 (1-\tau)E\{(1-\lambda_i^{**}) \\ & \cdot |\kappa_{jkl}(Z^{**(i)} + Z^* + \tau Z_i^{**})| \cdot |\dot{Z}_{ij}^{**} Z_{ik}^{**} Z_{il}^{**}|\}d\tau dt \\ \le& \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 (1-\tau)E\{(1-\lambda_i^{**}) \\ & \cdot Q_{jkl}(Z^{**(i)} + Z^* + \tau Z_i^{**}) \cdot |\dot{Z}_{ij}^{**} Z_{ik}^{**} Z_{il}^{**}|\}d\tau dt \\ \lesssim& \phi\beta^2 \sum_{i=1}^{m}\int_0^1 E\big\{\big(1-\lambda_i^{**}\big)\big(\max_{1\le j,k,l\le p}|\dot{Z}_{ij}^{**} Z_{ik}^{**} Z_{il}^{**}|\big)\big\}dt, \end{aligned} \tag{88}$$

where the first inequality is by Jensen's inequality, and the second and the third inequalities are based on Holder's inequality and Lemma 1. Note that for every $i = 1, \ldots, m$, we have

$$\begin{aligned} 1-\lambda_i^{**} =& 1\big\{\max_{1\le j\le p}|Y_{ij} - \mu_j^Y| \vee |G_{ij} - \mu_j^Y| \vee |W_{ij} - \mu_j^Y| > m^{1/2}/(4\beta|\delta_{n,m}|)\big\} \\ \le& 1\big\{\max_{1\le j\le p}|Y_{ij} - \mu_j^Y| > m^{1/2}/(4\beta|\delta_{n,m}|)\big\} + \\ & 1\big\{\max_{1\le j\le p}|G_{ij} - \mu_j^Y| > m^{1/2}/(4\beta|\delta_{n,m}|)\big\} + \\ & 1\big\{\max_{1\le j\le p}|W_{ij} - \mu_j^Y| > m^{1/2}/(4\beta|\delta_{n,m}|)\big\}. \end{aligned} \tag{89}$$

We also notice that for every $i = 1, \ldots, n$,

$$
\begin{aligned}
&\max_{1\le j,k,l\le p} |\dot{Z}^{**}_{ij} Z^{**}_{ik} Z^{**}_{il}| \\
\le& 100 m^{-3/2} |\delta_{n,m}|^3 \{t^{1/2} \wedge (1-t)^{1/2}\}^{-1} \\
&\cdot \max_{1\le j\le p} |Y_{ij} - \mu^Y_j|^3 \vee |G_{ij} - \mu^Y_j|^3 \vee |W_{ij} - \mu^Y_j|^3 \\
\le& 100 m^{-3/2} |\delta_{n,m}|^3 \{t^{1/2} \wedge (1-t)^{1/2}\}^{-1} \\
&\cdot \big(\max_{1\le j\le p} |Y_{ij} - \mu^Y_j|^3 + \max_{1\le j\le p} |G_{ij} - \mu^Y_j|^3 + \max_{1\le j\le p} |W_{ij} - \mu^Y_j|^3 \big). \qquad (90)
\end{aligned}
$$

Therefore, under (88), (89) and (90), we have

$$
\begin{aligned}
&|III^{**}_2| \\
\lesssim& m^{-3/2} \phi \beta^2 |\delta_{n,m}|^3 \sum_{i=1}^m E\Big(\Big[1\big\{ \max_{1\le j\le p} |Y_{ij} - \mu^Y_j| > m^{1/2}/(4\beta|\delta_{n,m}|) \big\} \\
&+ 1\big\{ \max_{1\le j\le p} |G_{ij} - \mu^Y_j| > m^{1/2}/(4\beta|\delta_{n,m}|) \big\} \\
&+ 1\big\{ \max_{1\le j\le p} |W_{ij} - \mu^Y_j| > m^{1/2}/(4\beta|\delta_{n,m}|) \big\} \Big] \\
&\cdot \Big[\max_{1\le j\le p} |Y_{ij} - \mu^Y_j|^3 + \max_{1\le j\le p} |G_{ij} - \mu^Y_j|^3 + \max_{1\le j\le p} |W_{ij} - \mu^Y_j|^3 \Big] \Big) \\
\lesssim& m^{-1/2} \phi \beta^2 |\delta_{n,m}|^3 \big\{ M^Y_m(\phi) + M^G_m(\phi) + M^W_m(\phi) \big\} \\
\lesssim& m^{-1/2} \phi \beta^2 |\delta_{n,m}|^3 \big\{ M^Y_m(\phi) + M^G_m(\phi) \big\} \\
\lesssim& m^{-1/2} \phi \beta^2 |\delta_{n,m}|^3 M^*_m(\phi), \qquad (91)
\end{aligned}
$$

where the second inequality follows from Lemma B.1 in [3], and the third inequal-

ity is based on the fact that $M_m^G(\phi) = M_m^W(\phi)$. For III_1^{**}, we have

$$
\begin{aligned}
&|III_1^{**}| \\
\leq &\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 (1-\tau)E\{\lambda_i^{**}|\kappa_{jkl}(Z^{**(i)}+Z^*+\tau Z_i^{**})| \\
&\cdot|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|\}d\tau dt \\
\leq &\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 E\{\lambda_i^{**}|\kappa_{jkl}(Z^{**(i)}+Z^*+\tau Z_i^{**})| \\
&\cdot|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|\}d\tau dt \\
= &\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 E\{\lambda_i^{**}h(Z^{**(i)}+Z^*,1) \\
&\cdot|\kappa_{jkl}(Z^{**(i)}+Z^*+\tau Z_i^{**})|\cdot|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|\}d\tau dt \\
\lesssim &\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1\int_0^1 E\{\lambda_i^{**}h(Z^{**(i)}+Z^*,1)Q_{jkl}(Z^{**(i)}+Z^*) \\
&\cdot|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|\}d\tau dt \\
\lesssim &\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{h(Z^{**(i)}+Z^*,1)Q_{jkl}(Z^{**(i)}+Z^*) \\
&\cdot|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|\}dt \\
\lesssim &\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{h(Z^{**(i)}+Z^*,1)Q_{jkl}(Z^{**(i)}+Z^*)\} \\
&\cdot E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)dt \\
\lesssim &\Delta_1^{**}+\Delta_2^{**}, \qquad (92)
\end{aligned}
$$

where the third inequality follows from Lemma 1, and the terms Δ_1^{**} and Δ_2^{**} are

$$
\begin{aligned}
\Delta_1^{**} = &\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{(1-\lambda_i^{**})h(Z^{**(i)}+Z^*,1) \\
&\cdot Q_{jkl}(Z^{**(i)}+Z^*)\}E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)dt, \\
\Delta_2^{**} = &\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\lambda_i^{**}h(Z^{**(i)}+Z^*,1) \\
&\cdot Q_{jkl}(Z^{**(i)}+Z^*)\}E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)dt.
\end{aligned}
$$

For Δ_1^{**}, it can be deduced that

$$
\begin{aligned}
&\Delta_1^{**}\\
=&\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{(1-\lambda_i^{**})h(Z^{**(i)}+Z^*,1)Q_{jkl}(Z^{**(i)}+Z^*)\}\\
&\cdot E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)dt\\
\le&\sum_{i=1}^{m}\int_0^1\Big\{\max_{1\le j,k,l\le p}E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)\Big\}\sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}E\big\{(1-\lambda_i^{**})\\
&\cdot h(Z^{**(i)}+Z^*,1)Q_{jkl}(Z^{**(i)}+Z^*)\big\}dt\\
\lesssim&\phi\beta^2\sum_{i=1}^{m}\int_0^1\Big\{\max_{1\le j,k,l\le p}E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)\Big\}E\big\{(1-\lambda_i^{**})\\
&\cdot h(Z^{**(i)}+Z^*,1)\big\}dt\\
\lesssim&\phi\beta^2\sum_{i=1}^{m}\int_0^1\Big\{\max_{1\le j,k,l\le p}E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)\Big\}\big\{E(1-\lambda_i^{**})\big\}dt\\
\lesssim&m^{-1/2}\phi\beta^2|\delta_{n,m}|^3\Big[\int_0^1\{t^{1/2}\wedge(1-t)^{1/2}\}^{-1}dt\Big]\\
&\cdot\big\{M_m^Y(\phi)+M_m^G(\phi)+M_m^W(\phi)\big\}\\
\lesssim&m^{-1/2}\phi\beta^2|\delta_{n,m}|^3\big\{M_m^Y(\phi)+M_m^G(\phi)\big\}\\
\lesssim&m^{-1/2}\phi\beta^2|\delta_{n,m}|^3M_m^*(\phi),
\end{aligned}
\tag{93}
$$

where the second inequality is based on Lemma 1, and the fourth inequality follows

from (89), (90) and Lemma B.1 in [3]. For Δ_2^{**}, it is clear that

$$\begin{aligned}
\Delta_2^{**} &= \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\lambda_i^{**}h(Z^{**(i)}+Z^*,1)Q_{jkl}(Z^{**(i)}+Z^*)\} \\
&\quad \cdot E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)dt \\
&\lesssim \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\lambda_i^{**}h(Z^{**(i)}+Z^*,1)Q_{jkl}(Z^{**}+Z^*)\} \\
&\quad \cdot E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)dt \\
&\lesssim \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{\lambda_i^{**}h(Z^{**(i)}+Z^*,1)h(Z^*+Z^{**},2) \\
&\quad \cdot Q_{jkl}(Z^{**}+Z^*)\}E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)dt \\
&\lesssim \sum_{j=1}^{p}\sum_{k=1}^{p}\sum_{l=1}^{p}\sum_{i=1}^{m}\int_0^1 E\{h(Z^*+Z^{**},2)Q_{jkl}(Z^{**}+Z^*)\} \\
&\quad \cdot E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)dt \\
&\lesssim \phi\beta^2\int_0^1 E\{h(Z^*+Z^{**},2)\}\big\{\max_{1\le j,k,l\le p}\sum_{i=1}^{m}E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)\big\}dt, \qquad (94)
\end{aligned}$$

where the first inequality follows from Lemma 1, the second inequality is based on the fact that

$$\lambda_i^{**}h(Z^{**(i)}+Z^*,1) = \lambda_i^{**}h(Z^{**(i)}+Z^*,1)h(Z^*+Z^{**},2), \quad i=1,\ldots,m,$$

and the fourth inequality follows from Holder's inequality and Lemma 1. Moreover, it is obvious that

$$\begin{aligned}
&\max_{1\le j,k,l\le p}\sum_{i=1}^{m}E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|) \\
\lesssim& m^{-3/2}|\delta_{n,m}|^3\{t^{1/2}\wedge(1-t)^{1/2}\}^{-1} \\
&\cdot \max_{1\le j\le p}\sum_{i=1}^{m}E(|Y_{ij}-\mu_j^Y|^3+|G_{ij}-\mu_j^Y|^3+|W_{ij}-\mu_j^Y|^3) \\
\lesssim& m^{-3/2}|\delta_{n,m}|^3\{t^{1/2}\wedge(1-t)^{1/2}\}^{-1} \\
&\cdot \big\{\max_{1\le j\le p}\sum_{i=1}^{m}E(|Y_{ij}-\mu_j^Y|^3)+\max_{1\le j\le p}\sum_{i=1}^{m}E(|G_{ij}-\mu_j^Y|^3)\big\}.
\end{aligned}$$

Together with the fact that $E(|G_{ij}-\mu_j^Y|^3) \lesssim \{E(|G_{ij}-\mu_j^Y|^2)\}^{3/2} \lesssim \{E(|Y_{ij}-\mu_j^Y|^2)\}^{3/2} \lesssim E(|Y_{ij}-\mu_j^Y|^3)$, we have

$$\max_{1\le j,k,l\le p}\sum_{i=1}^{m} E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|) \lesssim m^{-1/2}|\delta_{n,m}|^3\{t^{1/2}\wedge(1-t)^{1/2}\}^{-1}L_m^Y. \tag{95}$$

Thus, based on (78), (94) and (95), it is apparent that

$$\begin{aligned}\Delta_2^{**} &\lesssim \phi\beta^2\int_0^1 E\{h(Z^*+Z^{**},2)\}\big\{\max_{1\le j,k,l\le p}\sum_{i=1}^{m} E(|\dot{Z}_{ij}^{**}Z_{ik}^{**}Z_{il}^{**}|)\big\}dt\\ &\lesssim m^{-1/2}\phi\beta^2|\delta_{n,m}|^3\big\{L_m^Y\rho_{n,m}+\phi^{-1}L_m^Y(\log p)^{1/2}\big\}. \qquad (96)\end{aligned}$$

Then, according to (87), (91), (92), (93) and (96), one has

$$\begin{aligned}|III^{**}| \lesssim& m^{-1/2}\phi\beta^2|\delta_{n,m}|^3M_m^*(\phi)+\\ & m^{-1/2}\phi\beta^2|\delta_{n,m}|^3\big\{L_m^Y\rho_{n,m}+\phi^{-1}L_m^Y(\log p)^{1/2}\big\}\\ \lesssim& m^{-1/2}\beta^2|\delta_{n,m}|^3\big\{\phi L_m^Y\rho_{n,m}+L_m^Y(\log p)^{1/2}+\phi M_m^*(\phi)\big\}\\ \lesssim& m^{-1/2}\phi^2(\log p)^2|\delta_{n,m}|^3\big\{\phi L_m^Y\rho_{n,m}+L_m^Y(\log p)^{1/2}+\phi M_m^*(\phi)\big\}\\ \lesssim& m^{-1/2}\phi_2^2(\log p)^2|\delta_{n,m}|^3\big\{\phi_2 L_m^Y\rho_{n,m}+L_m^Y(\log p)^{1/2}+\phi_2 M_m^*(\phi_2)\big\}, \qquad (97)\end{aligned}$$

where the last inequality is based on the facts that $\phi_2\ge\phi$ and $M_m^*(\phi_2)\ge M_m^*(\phi)$. Thus, by combining (82), (83), (86) with (97), we have

$$\begin{aligned}|E(\mathcal{I}_{n,m}^{**})| \lesssim& m^{-1/2}\phi_2^2(\log p)^2|\delta_{n,m}|^3\\ &\cdot\big\{\phi_2 L_m^Y\rho_{n,m}+L_m^Y(\log p)^{1/2}+\phi_2 M_m^*(\phi_2)\big\}. \qquad (98)\end{aligned}$$

To this end, by combining (55), (81) with (98), we conclude that

$$\begin{aligned}|E(\mathcal{I}_{n,m})| \lesssim& n^{-1/2}\phi_1^2(\log p)^2\big\{\phi_1 L_n^X\rho_{n,m}+L_n^X(\log p)^{1/2}+\phi_1 M_n(\phi_1)\big\}+\\ & m^{-1/2}\phi_2^2(\log p)^2|\delta_{n,m}|^3\big\{\phi_2 L_m^Y\rho_{n,m}+L_m^Y(\log p)^{1/2}+\phi_2 M_m^*(\phi_2)\big\}. \qquad (99)\end{aligned}$$

Moreover, we have

$$
\begin{aligned}
& P\{Z(1) \le y - \phi^{-1}\} \\
\le & P\{F_\beta(Z(1)) \le 0\} \le E\{\kappa(Z(1))\} = E\{\kappa(Z(0))\} + \\
& E\{\kappa(Z(1)) - \kappa(Z(0))\} \\
= & E\{\kappa(Z(0))\} + E(\mathcal{I}_{n,m}) \le P\{F_\beta(Z(0)) \le \phi^{-1}\} + \\
& E(\mathcal{I}_{n,m}) \\
\le & P\{F_\beta(Z(0)) \le \phi^{-1}\} + |E(\mathcal{I}_{n,m})| \le P\{Z(0) \le y + \phi^{-1}\} + \\
& |E(\mathcal{I}_{n,m})| \\
= & P\{Z(0) \le y - \phi^{-1}\} + \big[P\{Z(0) \le y + \phi^{-1}\} - \\
& P\{Z(0) \le y - \phi^{-1}\}\big] + |E(\mathcal{I}_{n,m})| \\
\le & P\{Z(0) \le y - \phi^{-1}\} + c_1 \phi^{-1} (\log p)^{-1/2} + |E(\mathcal{I}_{n,m})| \qquad (100)
\end{aligned}
$$

where $c_1 > 0$ is a universal constant that depends only on b defined in condition (a), the first inequality is by (3), the second inequality is based on the fact that $1\{F_\beta(Z(1)) \le 0\} \le \kappa(Z(1))$, the third inequality is due to the fact that $1\{F_\beta(Z(0)) > \phi^{-1}\} \le 1 - \kappa(Z(0))$, the fifth inequality is owing to (3), and the last inequality follows from combining condition (a) with Lemma A.1 in [3]. Likewise, we have

$$
\begin{aligned}
& 1 - P\{Z(1) \le y + \phi^{-1}\} \\
\le & P\{F_\beta(Z(1)) > \phi^{-1}\} \le 1 - E\{\kappa(Z(1))\} = 1 - E\{\kappa(Z(0))\} - \\
& E\{\kappa(Z(1)) - \kappa(Z(0))\} \\
= & 1 - E\{\kappa(Z(0))\} - E(\mathcal{I}_{n,m}) \le 1 - P\{Z(0) \le y - \phi^{-1}\} - E(\mathcal{I}_{n,m}) \\
= & 1 - P\{Z(0) \le y + \phi^{-1}\} + \big[P\{Z(0) \le y + \phi^{-1}\} - \\
& P\{Z(0) \le y - \phi^{-1}\}\big] - E(\mathcal{I}_{n,m}) \\
\le & 1 - P\{Z(0) \le y + \phi^{-1}\} + c_1 \phi^{-1} (\log p)^{-1/2} + |E(\mathcal{I}_{n,m})|. \qquad (101)
\end{aligned}
$$

Finally, by combining (99), (100) with (101), we have

$$
\begin{aligned}
\rho_{n,m} \le & c_1 \phi^{-1} (\log p)^{-1/2} + \sup_{v \in [0,1]} \sup_{y \in \mathbb{R}^p} |E(\mathcal{I}_{n,m})| \lesssim (\min\{\phi_1, \phi_2\})^{-1} (\log p)^{1/2} \\
& + n^{-1/2} \phi_1^2 (\log p)^2 \big\{ \phi_1 L_n^X \rho_{n,m} + L_n^X (\log p)^{1/2} + \phi_1 M_n(\phi_1) \big\} \\
& + m^{-1/2} \phi_2^2 (\log p)^2 |\delta_{n,m}|^3 \big\{ \phi_2 L_m^Y \rho_{n,m} + L_m^Y (\log p)^{1/2} + \phi_2 M_m^*(\phi_2) \big\},
\end{aligned}
$$

which completes the proof. □

Proof of Lemma 3: First of all, given the data set $X^n = \{X_1, \ldots, X_n\}$, we construct another data set $\tilde{X}^n = \{\tilde{X}_1, \ldots, \tilde{X}_n\}$ with each $\tilde{X}_i = (\tilde{X}_{i1}, \ldots, \tilde{X}_{i,2p})' \in$

$\mathbb{R}^{2p}$ such that $\tilde{X}_{ij} = X_{ij}$ for $j = 1, \ldots, p$ and $\tilde{X}_{ij} = -X_{i,j-p}$ for $j = p+1, \ldots, 2p$. We further denote $E(\tilde{X}_i) = \mu^{\tilde{X}}$ for all $i = 1, \ldots, n$. Likewise, we also construct the data sets $\tilde{F}^n$, $\tilde{Y}^m$ and $\tilde{G}^m$. Based on these newly constructed data sets, we define the corresponding quantities $S_n^{\tilde{X}}$, $S_n^{\tilde{F}}$, $S_m^{\tilde{Y}}$, $S_m^{\tilde{G}}$, $L_n^{\tilde{X}}$, $L_n^{\tilde{F}}$, $L_m^{\tilde{Y}}$, $L_m^{\tilde{G}}$, $M_n^{\tilde{X}}(\phi)$, $M_n^{\tilde{F}}(\phi)$, $M_m^{\tilde{Y}}(\phi)$, $M_m^{\tilde{G}}(\phi)$, $\tilde{M}_n(\phi) = M_n^{\tilde{X}}(\phi) + M_n^{\tilde{F}}(\phi)$, $\tilde{M}_m^*(\phi) = M_m^{\tilde{Y}}(\phi) + M_m^{\tilde{G}}(\phi)$ analogously. Also denote

$$\begin{aligned}\tilde{\rho}_{n,m} = \sup_{v\in[0,1]} \sup_{\tilde{y}\in\mathbb{R}^{2p}} \big| & P\big\{v^{1/2}(S_n^{\tilde{X}} + \delta_{n,m} S_m^{\tilde{Y}} - n^{1/2}\mu^{\tilde{X}} - \delta_{n,m} m^{1/2}\mu^{\tilde{Y}}) + \\ & (1-v)^{1/2}(S_n^{\tilde{F}} + \delta_{n,m} S_m^{\tilde{G}} - n^{1/2}\mu^{\tilde{X}} - \delta_{n,m} m^{1/2}\mu^{\tilde{Y}}) \le \tilde{y}\big\} - \\ & P(S_n^{\tilde{F}} + \delta_{n,m} S_m^{\tilde{G}} - n^{1/2}\mu^{\tilde{X}} - \delta_{n,m} m^{1/2}\mu^{\tilde{Y}} \le \tilde{y})\big|.\end{aligned}$$

It is not difficult to verify that

$$\begin{aligned}& \min_{1\le j\le 2p} E\{(S_{nj}^{\tilde{X}} - n^{1/2}\mu_j^{\tilde{X}} + \delta_{n,m} S_{mj}^{\tilde{Y}} - \delta_{n,m} m^{1/2}\mu_j^{\tilde{Y}})^2\} \\ & = \min_{1\le j\le p} E\{(S_{nj}^{X} - n^{1/2}\mu_j^{X} + \delta_{n,m} S_{mj}^{Y} - \delta_{n,m} m^{1/2}\mu_j^{Y})^2\},\end{aligned} \tag{102}$$

and

$$L_n^{\tilde{X}} = L_n^X, \quad L_m^{\tilde{Y}} = L_m^Y, \quad \tilde{M}_n(\phi) = M_n(\phi), \quad \tilde{M}_m^*(\phi) = M_m^*(\phi), \tag{103}$$

for any $\phi \ge 1$. Given any $A \in \mathcal{A}^{Re}$ with the expression $A = \{w \in \mathbb{R}^p : a_j \le w_j \le b_j \quad \text{for all} \quad j = 1, \ldots, p\}$, we let $\tilde{y} \in (\tilde{y}_1, \ldots, \tilde{y}_{2p})' \in \mathbb{R}^{2p}$ such that $\tilde{y}_j = b_j$ for $j = 1, \ldots, p$ and $\tilde{y}_j = -a_{j-p}$ for $j = p+1, \ldots, 2p$. Then, it can be deduced that for any $v \in [0,1]$,

$$\begin{aligned}& \big| P\big\{v^{1/2}(S_n^X + \delta_{n,m} S_m^Y - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y) + \\ & (1-v)^{1/2}(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y) \in A\big\} - \\ & P(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y \in A)\big| \\ = & \big| P\big\{v^{1/2}(S_n^{\tilde{X}} + \delta_{n,m} S_m^{\tilde{Y}} - n^{1/2}\mu^{\tilde{X}} - \delta_{n,m} m^{1/2}\mu^{\tilde{Y}}) + \\ & (1-v)^{1/2}(S_n^{\tilde{F}} + \delta_{n,m} S_m^{\tilde{G}} - n^{1/2}\mu^{\tilde{X}} - \delta_{n,m} m^{1/2}\mu^{\tilde{Y}}) \le \tilde{y}\big\} - \\ & P(S_n^{\tilde{F}} + \delta_{n,m} S_m^{\tilde{G}} - n^{1/2}\mu^{\tilde{X}} - \delta_{n,m} m^{1/2}\mu^{\tilde{Y}} \le \tilde{y})\big|.\end{aligned}$$

This entails that

$$\begin{aligned}
&\rho^*_{n,m}\\
=&\sup_{v\in[0,1]}\sup_{A\in\mathcal{A}^{Re}}\big|P\big\{v^{1/2}(S^X_n+\delta_{n,m}S^Y_m-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y)+\\
&(1-v)^{1/2}(S^F_n+\delta_{n,m}S^G_m-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y)\in A\big\}-\\
&P(S^F_n+\delta_{n,m}S^G_m-n^{1/2}\mu^X-\delta_{n,m}m^{1/2}\mu^Y\in A)\big|\\
=&\sup_{v\in[0,1]}\sup_{\tilde{y}\in\mathbb{R}^{2p}}\big|P\big\{v^{1/2}(S^{\tilde{X}}_n+\delta_{n,m}S^{\tilde{Y}}_m-n^{1/2}\mu^{\tilde{X}}-\delta_{n,m}m^{1/2}\mu^{\tilde{Y}})+\\
&(1-v)^{1/2}(S^{\tilde{F}}_n+\delta_{n,m}S^{\tilde{G}}_m-n^{1/2}\mu^{\tilde{X}}-\delta_{n,m}m^{1/2}\mu^{\tilde{Y}})\le\tilde{y}\big\}-\\
&P(S^{\tilde{F}}_n+\delta_{n,m}S^{\tilde{G}}_m-n^{1/2}\mu^{\tilde{X}}-\delta_{n,m}m^{1/2}\mu^{\tilde{Y}}\le\tilde{y})\big|\\
=&\tilde{\rho}_{n,m}. \qquad (104)
\end{aligned}$$

In addition, by combining condition (a), (102) with Lemma 2, we have

$$\begin{aligned}
\tilde{\rho}_{n,m}\lesssim&n^{-1/2}\phi_1^2\{\log(2p)\}^2\big[\phi_1L^{\tilde{X}}_n\tilde{\rho}_{n,m}+L^{\tilde{X}}_n\{\log(2p)\}^{1/2}+\phi_1\tilde{M}_n(\phi_1)\big]+\\
&m^{-1/2}\phi_2^2\{\log(2p)\}^2|\delta_{n,m}|^3\\
&\cdot\big[\phi_2L^{\tilde{Y}}_m\tilde{\rho}_{n,m}+L^{\tilde{Y}}_m\{\log(2p)\}^{1/2}+\phi_2\tilde{M}^*_m(\phi_2)\big]\\
&+(\min\{\phi_1,\phi_2\})^{-1}\{\log(2p)\}^{1/2}. \qquad (105)
\end{aligned}$$

Hence by combining (103), (104), with (105), we conclude that

$$\begin{aligned}
\rho^*_{n,m}\le&K^*\Big[n^{-1/2}\phi_1^2(\log p)^2\big\{\phi_1L^X_n\rho^*_{n,m}+L^X_n(\log p)^{1/2}+\phi_1M_n(\phi_1)\big\}+\\
&m^{-1/2}\phi_2^2(\log p)^2|\delta_{n,m}|^3\big\{\phi_2L^Y_m\rho^*_{n,m}+L^Y_m(\log p)^{1/2}+\phi_2M^*_m(\phi_2)\big\}+\\
&(\min\{\phi_1,\phi_2\})^{-1}(\log p)^{1/2}\Big],
\end{aligned}$$

for a universal constant $K^*>0$ that depends only on b, which completes the proof. □

Proof of Lemma 4: First, we have

$$\rho^{**}_{n,m}\le\rho^*_{n,m}, \qquad (106)$$

by definition. Second, condition (a) together with Lemma 3 implies that for every $\phi_1,\phi_2\ge1$,

$$\begin{aligned}
\rho^*_{n,m}\le&K^*\Big[8n^{-1/2}\phi_1^2(\log p)^2\big\{\phi_1L^X_n\rho^*_{n,m}+2^{-1}L^X_n(\log p)^{1/2}+\phi_1M_n(2\phi_1)\big\}+\\
&8m^{-1/2}\phi_2^2(\log p)^2|\delta_{n,m}|^3\big\{\phi_2L^Y_m\rho^*_{n,m}+2^{-1}L^Y_m(\log p)^{1/2}+\phi_2M^*_m(2\phi_2)\big\}\\
&+(2\phi_1)^{-1}(\log p)^{1/2}+(2\phi_2)^{-1}(\log p)^{1/2}\Big], \qquad (107)
\end{aligned}$$

for the universal constant $K^* > 0$ defined in Lemma 3. Upon condition (b), we substitute $\phi_1 = 2^{-1}\phi_n^*$ and $\phi_2 = 2^{-1}\phi_m^{**}$ into (107) to obtain

$$
\begin{aligned}
\rho_{n,m}^* \leq & K^* \Big\{ n^{-1/2}(\phi_n^*)^3(\log p)^2 L_n^X \rho_{n,m}^* + n^{-1/2}(\phi_n^*)^2(\log p)^{5/2} L_n^X + \\
& n^{-1/2}(\phi_n^*)^3(\log p)^2 M_n(\phi_n^*) + m^{-1/2}(\phi_m^{**})^3(\log p)^2 L_m^Y |\delta_{n,m}|^3 \rho_{n,m}^* + \\
& m^{-1/2}(\phi_m^{**})^2(\log p)^{5/2} L_m^Y |\delta_{n,m}|^3 + m^{-1/2}(\phi_m^{**})^3(\log p)^2 |\delta_{n,m}|^3 M_m^*(\phi_m^{**}) \\
& + (\phi_n^*)^{-1}(\log p)^{1/2} + (\phi_m^{**})^{-1}(\log p)^{1/2} \Big\} \\
\leq & K^* \Big\{ n^{-1/2}(\phi_n^*)^3(\log p)^2 \bar{L}_n^* \rho_{n,m}^* + n^{-1/2}(\phi_n^*)^2(\log p)^{5/2} \bar{L}_n^* + \\
& n^{-1/2}(\phi_n^*)^3(\log p)^2 M_n(\phi_n^*) + m^{-1/2}(\phi_m^{**})^3(\log p)^2 \bar{L}_m^{**} |\delta_{n,m}|^3 \rho_{n,m}^* + \\
& m^{-1/2}(\phi_m^{**})^2(\log p)^{5/2} \bar{L}_m^{**} |\delta_{n,m}|^3 + m^{-1/2}(\phi_m^{**})^3(\log p)^2 |\delta_{n,m}|^3 M_m^*(\phi_m^{**}) \\
& + (\phi_n^*)^{-1}(\log p)^{1/2} + (\phi_m^{**})^{-1}(\log p)^{1/2} \Big\} \\
= & K^* \Big[K_1^3 \rho_{n,m}^* + K_1^2 \{(\bar{L}_n^*)^2(\log p)^7/n\}^{1/6} + K_1^3 \{M_n(\phi_n^*)/\bar{L}_n^*\} + K_1^3 \rho_{n,m}^* \\
& + K_1^2 \{(\bar{L}_m^{**})^2(\log p)^7 |\delta_{n,m}|^6/m\}^{1/6} + K_1^3 \{M_m^*(\phi_m^{**})/\bar{L}_m^{**}\} + \\
& K_1^{-1} \{(\bar{L}_n^*)^2(\log p)^7/n\}^{1/6} + K_1^{-1} \{(\bar{L}_m^{**})^2(\log p)^7 |\delta_{n,m}|^6/m\}^{1/6} \Big] \\
= & 2K^* K_1^3 \rho_{n,m}^* + K^*(K_1^2 + K_1^{-1}) \{(\bar{L}_n^*)^2(\log p)^7/n\}^{1/6} + \\
& K^* K_1^3 \{M_n(\phi_n^*)/\bar{L}_n^*\} + K^*(K_1^2 + K_1^{-1}) \{(\bar{L}_m^{**})^2(\log p)^7 |\delta_{n,m}|^6/m\}^{1/6} \\
& + K^* K_1^3 \{M_m^*(\phi_m^{**})/\bar{L}_m^{**}\} \\
\leq & 2^{-1} \rho_{n,m}^* + K^*(K_1^2 + K_1^{-1}) \{(\bar{L}_n^*)^2(\log p)^7/n\}^{1/6} + 4^{-1} \{M_n(\phi_n^*)/\bar{L}_n^*\} \\
& + K^*(K_1^2 + K_1^{-1}) \{(\bar{L}_m^{**})^2(\log p)^7 |\delta_{n,m}|^6/m\}^{1/6} + 4^{-1} \{M_m^*(\phi_m^{**})/\bar{L}_m^{**}\},
\end{aligned} \tag{108}
$$

where the last inequality is by $K_1 \in (0, (K^* \vee 2)^{-1}]$. It then follows from (108) that

$$
\begin{aligned}
\rho_{n,m}^* \leq & K_2 \big[\{(\bar{L}_n^*)^2(\log p)^7/n\}^{1/6} + \{M_n(\phi_n^*)/\bar{L}_n^*\} + \\
& \{(\bar{L}_m^{**})^2(\log p)^7 |\delta_{n,m}|^6/m\}^{1/6} + \{M_m^*(\phi_m^{**})/\bar{L}_m^{**}\} \big],
\end{aligned} \tag{109}
$$

where $K_2 = \max\{2K^*(K_1^2 + K_1^{-1}), 2^{-1}\}$ is a universal constant that depends on b only. Putting (106) and (109) together leads to

$$
\begin{aligned}
\rho_{n,m}^{**} \leq & K_2 \big[\{(\bar{L}_n^*)^2(\log p)^7/n\}^{1/6} + \{M_n(\phi_n^*)/\bar{L}_n^*\} + \\
& \{(\bar{L}_m^{**})^2(\log p)^7 |\delta_{n,m}|^6/m\}^{1/6} + \{M_m^*(\phi_m^{**})/\bar{L}_m^{**}\} \big],
\end{aligned}
$$

which completes the proof. □

Proof of Lemma 5: Given any $\phi > 0$, we denote $\beta = \phi \log p$. Given any $y \in \mathbb{R}^p$, we define the function $F_\beta(w)$ as in (2) that implicitly depends on y. Then, piecing $F_\beta(w)$ and ϕ together, we construct a function $\kappa(w) = \varphi_0(\phi F_\beta(w)) = \varphi(F_\beta(w))$ according to (4). For any sequence of constants $\delta_{n,m}$ that depends on both n and m, we denote $\rho_{n,m}^{MBB}$ by

$$\rho_{n,m}^{MBB} = \sup_{y\in\mathbb{R}^p} \big| P_e(S_n^{eX} + \delta_{n,m} S_m^{eY} \le y) - P(S_n^F - n^{1/2}\mu^X + \delta_{n,m} S_m^G - \delta_{n,m} m^{1/2}\mu^Y \le y) \big|,$$

where $P_e(\cdot)$ means the probability with respect to e^{n+m} only. Now we are in a position to start the proof. First, it is clear that

$$\begin{aligned}
&P_e(S_n^{eX} + \delta_{n,m} S_m^{eY} \le y - \phi^{-1}) \\
\le& P_e\big\{F_\beta(S_n^{eX} + \delta_{n,m} S_m^{eY}) \le 0\big\} \le E_e\big\{\kappa(S_n^{eX} + \delta_{n,m} S_m^{eY})\big\} \\
=& E\big\{\kappa(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y)\big\} + \big[E_e\big\{\kappa(S_n^{eX} + \delta_{n,m} S_m^{eY})\big\} \\
&- E\big\{\kappa(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y)\big\}\big] \\
\le& P\big\{F_\beta(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y) \le \phi^{-1}\big\} + \big[E_e\big\{\kappa(S_n^{eX} + \\
&\delta_{n,m} S_m^{eY})\big\} - E\big\{\kappa(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y)\big\}\big] \\
\le& P(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y \le y + \phi^{-1}) + \big[E_e\big\{\kappa(S_n^{eX} + \\
&\delta_{n,m} S_m^{eY})\big\} - E\big\{\kappa(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y)\big\}\big] \\
\le& P(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y \le y - \phi^{-1}) + c_1\phi^{-1}(\log p)^{1/2} + \\
&\big|E_e\big\{\kappa(S_n^{eX} + \delta_{n,m} S_m^{eY})\big\} - E\big\{\kappa(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y)\big\}\big|, \qquad (110)
\end{aligned}$$

where $c_1 > 0$ is a universal constant depending only on b, and the last inequality follows from combining condition (a) with Lemma A.1 in [3]. To bound the term $\big|E_e\big\{\kappa(S_n^{eX} + \delta_{n,m} S_m^{eY})\big\} - E\big\{\kappa(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y)\big\}\big|$ in (110), one has

$$\begin{aligned}
&\big|E_e\big\{\kappa(S_n^{eX} + \delta_{n,m} S_m^{eY})\big\} - E\big\{\kappa(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y)\big\}\big| \\
=& \big|E_e\big\{\varphi(F_\beta(S_n^{eX} + \delta_{n,m} S_m^{eY}))\big\} - \\
&E\big\{\varphi(F_\beta(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y))\big\}\big| \\
\le& (\|\varphi''\|_\infty/2 + \beta\|\varphi'\|_\infty)\hat{\Delta}_{n,m} \\
\lesssim& (\phi^2 + \beta\phi)\hat{\Delta}_{n,m} \qquad (111) \\
\lesssim& (\phi^2 \log p)\hat{\Delta}_{n,m},
\end{aligned}$$

where the first inequality follows from Theorem 1 in [2]. Then it follows from (110) and (111) that

$$\begin{aligned} &P_e(S_n^{eX} + \delta_{n,m} S_m^{eY} \le y - \phi^{-1}) - \\ &P(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y \le y - \phi^{-1}) \\ \lesssim &\phi^{-1}(\log p)^{1/2} + (\phi^2 \log p)\hat{\Delta}_{n,m}. \end{aligned} \tag{112}$$

Similar argument leads to

$$\begin{aligned} &P(S_n^F + \delta_{n,m} S_m^G - n^{1/2}\mu^X - \delta_{n,m} m^{1/2}\mu^Y \le y - \phi^{-1}) \\ &- P_e(S_n^{eX} + \delta_{n,m} S_m^{eY} \le y - \phi^{-1}) \\ \lesssim &\phi^{-1}(\log p)^{1/2} + (\phi^2 \log p)\hat{\Delta}_{n,m}. \end{aligned} \tag{113}$$

As a consequence, piecing (112) and (113) together yields that

$$\rho_{n,m}^{MBB} \lesssim \phi^{-1}(\log p)^{1/2} + (\phi^2 \log p)\hat{\Delta}_{n,m}. \tag{114}$$

By substituting $\phi = 2^{-1/3}(\log p)^{-1/6}(\hat{\Delta}_{n,m})^{1/3}$ into (114), we obtain

$$\rho_{n,m}^{MBB} \lesssim (\log p)^{2/3}\hat{\Delta}_{n,m}.$$

Together with the same reasoning as in the proof of Lemma 3, we conclude that

$$\rho_{n,m}^{MB} \lesssim (\log p)^{2/3}\hat{\Delta}_{n,m},$$

which completes the proof. □

Proof of Lemma 6: First, condition (a) entails that

$$\lim_{k_1\to\infty} \lim_{k_2\to\infty} \sum_{n=k_1}^{\infty} \sum_{m\in\varrho(k_2)\cap\sigma(n)} P(A_{n,m}) = 0. \tag{115}$$

In addition, one has

$$\begin{aligned} P\Big(\bigcap_{k_1=1}^{\infty}\bigcap_{k_2=1}^{\infty}\bigcup_{n=k_1}^{\infty}\bigcup_{m\in\varrho(k_2)\cap\sigma(n)} A_{n,m}\Big) \le &P\Big(\bigcup_{n=k_1}^{\infty}\bigcup_{m\in\varrho(k_2)\cap\sigma(n)} A_{n,m}\Big) \\ &\le \sum_{n=k_1}^{\infty}\sum_{m\in\varrho(k_2)\cap\sigma(n)} P(A_{n,m}), \end{aligned} \tag{116}$$

for all $k_1, k_2 \ge 1$. Finally, by combining (115) with (116), we conclude that

$$\begin{aligned} P\Big(\bigcap_{k_1=1}^{\infty}\bigcap_{k_2=1}^{\infty}\bigcup_{n=k_1}^{\infty}\bigcup_{m\in\varrho(k_2)\cap\sigma(n)} A_{n,m}\Big) &\le \lim_{k_1\to\infty}\lim_{k_2\to\infty}\sum_{n=k_1}^{\infty}\sum_{m\in\varrho(k_2)\cap\sigma(n)} P(A_{n,m}) \\ &= 0. \end{aligned}$$

□

REFERENCES

[1] CHERNOZHUKOV, V., CHETVERIKOV, D. and KATO, K. (2013). Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. *The Annals of Statistics* **41** 2786–2819.

[2] CHERNOZHUKOV, V., CHETVERIKOV, D. and KATO, K. (2015). Comparison and anti-concentration bounds for maxima of Gaussian random vectors. *Probability Theory and Related Fields* **162** 47–70.

[3] CHERNOZHUKOV, V., CHETVERIKOV, D. and KATO, K. (2017). Central limit theorems and bootstrap in high dimensions. *The Annals of Probability* **45** 2309–2352.

[4] VAN DER VAART, A. and WELLNER, J. A. (1996). *Weak Convergence and Empirical Processes: With Applications to Statistics*. Springer.

K. XUE
SCHOOL OF STATISTICS AND DATA SCIENCE
NANKAI UNIVERSITY
TIANJIN, CHINA
E-MAIL: kaijie@nankai.edu.cn

F. YAO
CORRESPONDING AUTHOR
DEPARTMENT OF PROBABILITY AND STATISTICS
CENTER FOR STATISTICAL SCIENCE
SCHOOL OF MATHEMATICAL SCIENCES
PEKING UNIVERSITY
BEIJING, CHINA
E-MAIL: fyao@math.pku.edu.cn